The Limits of the Expectations Hypothesis

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Abstract

Departures from the expectations hypothesis (EH) at the very short end of the US repo curve are statistically detectable but worth at most a few basis points a year. VAR-based tests reject the EH there, but they assume a linear model of expectations, and their asymptotic Wald version rejects a true EH in at least 86% of samples simulated from a stylised model. We test an implication that requires no parametric model of expectations: under rational expectations and a constant premium, the excess return of rolling over short repos instead of holding a longer repo is unpredictable. On daily general-collateral repo rates from overnight to one month, 1998–2026, linear, tree-based, and neural-network forecasts compete out of sample under one protocol. Only at horizons of one and two weeks does predictability remain significant after accounting for the search over forecasts, and for the one- versus two-week pair mainly in the zero-lower-bound years. Linear forecasts capture it as well as any flexible forecast. For the overnight versus one- and two-week pairs, about three quarters of it disappears when same-day composite quotes are excluded. None of the 22 forecasts earns more than 2.22 basis points a year before costs or 0.86 after observed bid–ask spreads. Even perfect foresight of the sign of the excess return would earn at most 4.15 basis points a year per unit of notional before costs.

Keywords: Expectations hypothesis, Term premium, Repo rates, Forecast evaluation, Economic value, Machine learning
JEL: E43, G12, C53

1. Introduction

A cash investor can lock in a one-week repo rate today or roll over overnight repos for a week. Under the expectations hypothesis (EH) the choice does not matter apart from a constant premium. Longstaff (2000) finds support for the EH in repo rates, and Zhu (2014) attributes many of the remaining anomalies to peso problems. Della Corte et al. (2008) (DST) reject the EH for every pair of US repo maturities they test, from overnight to three

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months. Yet trading on the rejections earns at most 1.34 basis points a year before costs at a 1% volatility target, and its value after costs is not economically significant. They caution that their tests rely on a linear vector autoregression (VAR) as the model of expectations. Standard EH tests are joint tests of the hypothesis and of how expectations are formed (Guidolin and Thornton, 2018; Crump et al., 2024). A nonlinear process could therefore create spurious rejections or hide economically important departures.

We ask whether economically meaningful failure of the EH survives without the linear expectations model. We test a restriction that requires no parametric model of expectations, only rational expectations and a constant premium: the excess return of rolling over short repos instead of holding the long repo is unpredictable with information available at the investment date. Its best forecast is a constant, estimated in real time by the prevailing mean; any model that beats it out of sample is evidence against the EH. Linear and flexible forecasts compete under one protocol. In-sample tests cannot settle the question: an asymptotic Wald test of DST's VAR restrictions rejects a true EH in at least 86% of simulated samples.

Three results emerge (Figures 1 and 2). First, predictability that survives the search over forecasts is confined to horizons of one and two weeks, for 1W-2W mainly to the zero-lower-bound years, and linear forecasts capture it as well as any flexible forecast. Second, for ON-1W and ON-2W about three quarters of this predictability is tied to same-day composite quotes. Third, the failure is economically bounded: none of the 22 forecasts earns more than 2.22 basis points a year before costs or 0.86 after observed bid–ask spreads, and even perfect sign foresight earns at most 4.15 basis points a year per unit of notional before costs.

[Insert Figure 1 here]

These results strengthen DST's economic verdict. DST evaluate one linear model in sample on 1991–2005 data; our evaluation is out of sample to 2026, needs no parametric model of expectations, and bounds the gross value of any forecast in a common trading strategy, because all forecasts take positions of the same size and differ only in sign. The letter adds to out-of-sample evidence on the EH (Guidolin and Timmermann, 2009; Thornton

and Valente, 2012; Sarno et al., 2016; Guidolin and Thornton, 2018), traces much of the short-end predictability to same-day quotes, and contrasts with machine-learning evidence of large bond-premium predictability at long maturities (Bianchi et al., 2021b,a): at the short end, flexible forecasts do not change the economic verdict.

2. Testing the EH Without a Parametric Model of Expectations

Let $i_t^{(n)}$ be the rate at date t on a repo with a maturity of n business days, and let the short repo have maturity m, with $n = km$ for an integer k. Repo rates are simple ACT/360 rates. Holding the long repo returns $r_t^f = i_t^{(n)} D_t/360$ over its life of D_t calendar days. Rolling the short repo k times returns $r_t = \prod_{j=0}^{k-1}(1 + i_{t_j}^{(m)} d_j/360) - 1$, where roll j starts on the business day $t_j = t + jm$ and lasts d_j calendar days, so that $\sum_j d_j = D_t$. The annualised excess return is

$$xr_t^{(n,m)} = \left(r_t - r_t^f\right) 360/D_t. \tag{1}$$

With a constant term premium $\pi^{(n,m)}$ and rational expectations, the EH implies

$$E_t\left[xr_t^{(n,m)}\right] = -\pi^{(n,m)}. \tag{2}$$

The deviation $xr_t^{(n,m)} + \pi^{(n,m)}$ is then orthogonal to any function of the information at t, linear or not. The VAR test of DST imposes the same hypothesis through cross-equation restrictions that are valid only if expectations follow the VAR.

The restriction maintains three assumptions: the premium is constant, expectations are rational given the information at t, and quoted rates are prices at which investors can trade. Section 4.1 relaxes the first by allowing a slowly moving premium and examines the third with transaction-based rates. A rejection means only that the excess return is predictable, through a time-varying premium, expectation errors, or quote noise; whether this is exploitable is measured separately.

The expected excess return need not be linear in the state. A lower bound on rates and

regime switches make it nonlinear in term-structure models (Bianchi et al., 2021b, Section 1), and so do policy-rate changes on meeting dates (Supplement, Section S7). Neural networks forecast one-year Treasury excess returns better than linear methods, even from yields alone, although less strongly after correction (Bianchi et al., 2021b,a). In our data, the RESET test rejects linearity for ON-2W and 1W-2W, and a neural-network test only for ON-2W (Supplement, Table S5).

3. Data and Forecast Design

We use Bloomberg composite closing quotes of US general-collateral Treasury repo rates for overnight (ON), one and two weeks (1W, 2W), and one month (1M), from 23 July 1998 to 30 September 2026. We do not use the 3W, 2M, and 3M composites, which have not changed since December 2015 while other repo rates rose above 5%. The four maturities give the pairs ON-1W, ON-2W, ON-1M, and 1W-2W. Even these series contain frozen periods (Supplement, Figure S1). We therefore set to missing all but the first quote of every run of identical values longer than 20 business days, which removes between 18% (ON) and 42% (2W) of the observations. The results are similar with thresholds of 10 or 40 days and with a rule that uses only past quotes (Supplement, Section S6). The out-of-sample forecasts start between 2006 and 2009 and extend to 2026.

Every model forecasts the excess return of Equation (1) on each business day t from five daily lags of the pair's two rates, the information set of DST's VAR(5). Rolling windows contain 1,250 business days and include only excess returns realised by t. Models are re-estimated every 21 forecast dates and tuned yearly on the last 15% of each window, which is separated from the training targets by the overlap of the returns. Supplement Table S1 documents this protocol. The EH forecast is the prevailing mean, and the VAR forecast is an ordinary least squares regression on the lags. The 19 machine-learning models are regularised linear, kernel, nearest-neighbour, tree-based, random-feature, and neural-network models with one to five layers. With the VAR, an equal-weighted combination of the individual

forecasts, and an exponentially weighted mean of past excess returns (EWMA), the 22 forecasts are evaluated on the same dates.

Accuracy is the out-of-sample R^2_{oos} relative to the EH forecast (Campbell and Thompson, 2008). Individual forecasts are tested with Clark and West (2007), using Newey–West variances with the automatic bandwidth of Andrews (1991), but at least as many lags as the horizon. Because the EH concerns every forecast, the test of superior predictive ability (SPA) of Hansen (2005) asks whether any of the 22 forecasts beats the EH. Economic value is the annual performance fee (Fleming et al., 2001) that an investor with relative risk aversion of 5 would pay to switch from the EH strategy to a forecast-based strategy, following DST. Positions are scaled to about one unit of notional rather than to DST's volatility targets, so fees are not directly comparable with DST's. Each change of position between a maturing and a new tranche pays a bid–ask spread of either 4 basis points, as in DST, or the observed spread of the pair, the average of the mean spreads of its two maturities; these range from 1 basis point (2W, 1M) to 6 (ON). To bound the gross value of any forecast in this strategy, we also evaluate an investor with perfect foresight of the sign of the excess return.

4. Results

[Insert Table 1 here]

4.1. What survives out of sample?

Panel A of Table 1 shows why an asymptotic in-sample test cannot settle whether the EH fails. A Wald test of DST's VAR restrictions rejects the EH for all four pairs. Yet it also rejects in 86% to 98% of samples simulated from a stylised two-factor model in which the EH holds up to a compounding term, in line with the size distortions documented by Bekaert and Hodrick (2001). Against the simulated distributions, no rejection in the data is significant (p-values 0.34 to 0.84). DST's bootstrap tests have better size but still impose the linear VAR.

Out of sample, the answer differs by horizon (Panel B). The VAR predicts the excess returns of ON-1W, ON-2W, and 1W-2W with R^2_{oos} of 42.2%, 26.4%, and 50.1%, and the term spread alone does about as well, so this is the classical predictability. For ON-1M, small individual gains do not survive the search over the 22 forecasts (SPA $p = 0.91$). The Clark–West test is oversized under the EH, but the VAR's R^2_{oos} exceeds every simulated value for the three predictable pairs (Supplement, Table S17). Predictability that survives the search over forecasts is thus confined to horizons of one and two weeks. The model confidence set (Hansen et al., 2011) of all forecasts, which includes very poor ones, is less decisive and retains the EH forecast for every pair (Supplement, Table S7).

Panel C shows what this predictability consists of. On the same dates, dropping the same-day quotes lowers the VAR's R^2_{oos} from 42.2% to 10.1% (ON-1W), from 26.5% to 6.2% (ON-2W), and from 50.0% to 28.6% (1W-2W). About three quarters of the predictability disappears for ON-1W and ON-2W, but less than half for 1W-2W. Noise in the same-day long-rate quote would enter the predictors and, with the opposite sign, the excess return; a short-rate error would enter only with a weight of about $1/k$ (Supplement, Section S7; Figure 2, Panel B). Transaction-based overnight rates show how noisy composite quotes are (Figure 2, Panel A): since 2018 the composite overnight rate changes on 89% of days, the transaction-based rates of the Federal Reserve Bank of New York on about half, and their daily changes correlate at only 0.40. A slowly moving premium explains part of the predictability: the EWMA attains R^2_{oos} of up to 18.1% (1W-2W), but the best forecasts beat it for all three pairs (SPA $p \leq 0.009$).

[Insert Figure 2 here]

The holdout from 21 April 2021 (Panel D) repeats this pattern for ON-1W and ON-2W. The VAR's R^2_{oos} is 61.3% (ON-1W) and 40.1% (ON-2W) with same-day quotes and 15.1% and 10.1% without them. For 1W-2W it is only 11.1%, and −3.2% without: this pair's predictability rests mainly on the zero-lower-bound years; away from them the VAR attains 21.0%, and 3.4% without same-day quotes (Supplement, Table S19). Models, grids, windows,

and tests were fixed on data to April 2021; the choice of maturities and the treatment of holidays were revised in 2026 on data-quality grounds. The gains survive the exclusion of the main stress episodes: without September 2008–June 2009, September–December 2019, March–June 2020, and year-ends, the VAR's R^2_{oos} stays between 32% and 58% for the three predictable pairs (Supplement, Table S17), and it remains above 16% on non-overlapping dates (Table S19). Without same-day quotes, what survives is small and imprecise: the Clark–West test still rejects for ON-1W and ON-2W, but the 95% bootstrap intervals of R^2_{oos} include zero (Table S19).

4.2. Does functional-form flexibility matter?

The flexible forecasts do not reveal predictability that the linear model misses. The best of the 19 machine-learning models falls short of the VAR for ON-1W, ON-2W, and 1W-2W (Table 1). With the same information in stationary form, flexible forecasts improve but exceed the VAR only for ON-2W, and not significantly after the search (SPA $p = 0.052$), and 100 macro-financial predictors similar to those of Ludvigson and Ng (2009) lower accuracy (Supplement, Table S20). None of 84 pairwise Diebold and Mariano (1995) comparisons with the VAR favours the alternative at the 5% level. After a Holm adjustment, machine-learning models add information to the VAR only for ON-2W, whose predictability is concentrated around month turns (Supplement, Tables S7 and S12).

Agreement between the models is not proof of linearity. In simulations with known departures, the flexible models detect a nonlinear signal only when it explains about 30% of the variance (Supplement, Section S5). Relaxing the functional form thus leaves the inference unchanged but cannot exclude sizeable nonlinear departures; the economic bound below does not need to.

[Insert Table 2 here]

4.3. Does the rejection matter economically?

Table 2 and Figure 1 translate predictability into economic value. Before costs, the VAR is worth 1.85 basis points a year for ON-1W and 2.22 for 1W-2W, and none of the 22 forecasts, even when chosen ex post, exceeds 2.22 on any pair. Costs then decide the sign: the VAR's break-even spread, 1.7 to 2.9 basis points, lies within the range of observed spreads. With DST's 4-basis-point spread, the VAR loses relative to the EH on every pair, and no forecast earns more than 0.72 basis points. With the observed spreads of each pair, the VAR earns 0.71 basis points for 1W-2W and loses on the other pairs, and no forecast earns more than 0.86.

The perfect-foresight columns of Table 2 bound the gross value of any forecast in this strategy. On average, the realised excess return deviates from the EH forecast by 3.5 to 6.1 basis points a year, whatever its source, including quote noise. The premium itself is small: on average, holding the long repo earned 0.2 to 3.4 basis points a year more than rolling (Supplement, Table S19). An investor who knows the sign of the excess return in advance earns 3.29 to 4.15 basis points a year per unit of notional before costs. On a cash book of USD 10 billion, that is at most USD 4.15 million a year; the best actual forecast earns at most USD 0.9 million after observed spreads. After costs the bound no longer applies, because perfect foresight trades whenever the sign changes; with DST's spread, the rarely trading EWMA earns most on every pair. Statistical predictability can thus be strong while its economic value, in this strategy, stays at a few basis points a year per unit of notional.

5. Conclusion

At the short end of the US repo curve, the EH fails statistically but is bounded economically. Out-of-sample predictability that survives the search over forecasts is confined to horizons of one and two weeks, for 1W-2W mainly to the zero-lower-bound years. Linear forecasts capture it as well as flexible ones, and for ON-1W and ON-2W it largely disappears without same-day quotes. Even perfect sign foresight would earn only a few basis points a

year before costs. The bound holds per unit of notional for a sign-based strategy, and the net results use average bid–ask spreads, which widen in stress. The evidence covers maturities up to one month, so three of the four pairs on which DST find small positive fees after costs cannot be tested. Transaction-based term rates and other currencies would test the bound further. For cash investors choosing between term and overnight repo, the EH remains a sound benchmark.

Declaration of competing interest

The author is employed by Quintet Private Bank (Europe) S.A.; the views expressed are the author's own and do not necessarily reflect those of Quintet Private Bank. The author declares no other known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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CRediT authorship contribution statement

Dennis Jung: Conceptualization, Data curation, Formal analysis, Methodology, Software, Validation, Visualization, Writing – original draft, Writing – review & editing.

Data availability

The repo quotes are proprietary Bloomberg data and cannot be shared under the data licence. The transaction-based rates (secured overnight financing rate, broad and tri-party general collateral rates; Federal Reserve Bank of New York) and the federal funds target (FRED) are publicly available. The code that reproduces all results from these data is available from the author on request.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the author used Claude (Anthropic) in order to support programming and text editing. After using this tool, the author reviewed and edited the content as needed and takes full responsibility for the content of the published article.

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Table 1: **What Survives Out of Sample**

	ON-1W	ON-2W	ON-1M	1W-2W
Panel A: In-sample Wald test of DST's VAR restrictions				
Wald test, asymptotic p-value	< 0.001	< 0.001	< 0.001	< 0.001
Wald test, p-value simulated under the EH	0.470	0.800	0.340	0.840
Rejection rate of a true EH at 5% (%)	86.0	94.5	96.0	97.5
Panel B: Out-of-sample R^2 relative to the EH forecast (%)				
Term spread only	36.9^{***}	26.2^{***}	1.7^{***}	46.7^{***}
VAR (DST information set)	42.2^{***}	26.4^{***}	-15.4	50.1^{***}
Ridge (regularised linear)	28.9^{***}	23.6^{***}	-13.5	48.6^{***}
Random forest (trees)	-58.6	-0.2^{***}	-54.6^{*}	31.3^{***}
Neural-network ensemble	3.6^{***}	1.2^{***}	-5.2	6.7^{***}
Best of 19 machine-learning models	28.9^{***}	26.3^{***}	1.4^{**}	48.6^{***}
SPA p-value: no model beats the EH	0.007	0.000	0.909	0.010
Panel C: What the predictability consists of				
VAR with same-day quotes, same dates, R^2_{oos} (%)	42.2	26.5	-15.5	50.0
VAR without same-day quotes, R^2_{oos} (%)	10.1^{***}	6.2^{***}	-18.9	28.6^{***}
Slowly moving premium (EWMA), R^2_{oos} (%)	1.1	7.7	-2.3	18.1
SPA p-value: no model beats the EWMA	0.009	0.002	0.817	0.003
Panel D: Holdout, 21 April 2021 to September 2026, R^2_{oos} (%)				
VAR	61.3^{***}	40.1^{***}	-46.5	11.1^{***}
VAR without same-day quotes	15.1^{***}	10.1^{***}	-56.3	-3.2^{**}
Out-of-sample observations	4,688	3,276	3,110	3,331

The target is the annualised excess return of rolling the short repo over holding the long repo, Equation (1); its forecast under the expectations hypothesis (EH) is the prevailing mean. Panel A: Wald test of the restrictions on a vector autoregression with five lags, VAR(5), of Della Corte et al. (2008) (DST) and its p-value and size in 200 samples simulated from a two-factor model in which the EH holds up to a compounding term (Supplement, Section S2). Panel B: R^2_{oos} of Campbell and Thompson (2008); stars (Panels B to D): Clark and West (2007) test that the EH forecast is as accurate as the model in population, at the 10%, 5%, and 1% level, Newey–West variance with the automatic bandwidth of Andrews (1991), but at least as many lags as the horizon; because the statistic adjusts for the noise of the competing forecast, it can reject even when the sample R^2_{oos} is negative. All models use five daily lags of the two rates, the same rolling five-year windows, and the same tuning protocol (Table S1). The best machine-learning model is the maximum over 19 models (regularised linear, kernel, tree-based, random-feature, and neural-network models); the test of superior predictive ability (SPA) of Hansen (2005) accounts for the search over all 22 forecasts. Panel C: the VAR with lags dated $t-1$ to $t-5$ instead of t to $t-4$, both evaluated on the dates on which both lag sets are available (hence the small differences from Panel B); an exponentially weighted mean (EWMA) of past excess returns (half-life six months); and the SPA test with this mean as the benchmark. Term spread only: regression on the term spread (long minus short rate) at t, the predictor of the classical tests, not one of the 22 forecasts. Panel D: forecasts made after the design was fixed on data to April 2021, with the stars of Panel B (Supplement, Table S19).

Table 2: **Economic Value of Departures from the EH and Its Upper Bound**

	Spread	VAR				Best of 22 forecasts			Perfect foresight of the sign				
	observed	Gross	Net 4 bp	Net obs.	Break-even	Gross	Net 4 bp	Net obs.	$\lvert xr - \overline{xr}\rvert$	Gross	Net 4 bp	Net obs.	Break-even
ON-1W	4.0	1.85	−1.37	−1.37	2.30	1.85	0.56	0.56	5.0	3.69	−0.13	−0.13	3.86
ON-2W	3.5	0.87	−1.23	−0.97	1.66	1.04	0.72	0.73	5.2	3.29	−0.01	0.40	3.99
ON-1M	3.5	0.41	−0.17	−0.09	2.86	0.99	0.71	0.73	6.1	3.60	1.24	1.53	6.10
1W-2W	1.5	2.22	−1.83	0.71	2.20	2.22	0.17	0.86	3.5	4.15	−0.63	2.35	3.47

Annualised performance fee $\mathcal{F}$ in basis points (bp) that an investor with quadratic utility and relative risk aversion of 5 would pay to switch from the strategy implied by the expectations hypothesis (EH; prevailing-mean forecast) to a strategy based on the forecasts (Fleming et al., 2001), as in Della Corte et al. (2008) (DST). VAR: the forecast from five lags of the two rates, DST's information set. The position in the rolling strategy is $w_t = (\bar{\sigma}_t/\widehat{\sigma}_t)\,\mathrm{sign}(\widehat{\mu}_t)$, where $\widehat{\mu}_t$ is the forecast, $\widehat{\sigma}_t$ the volatility of past excess returns, and $\bar{\sigma}_t$ its average over the estimation window, so the median position is about one unit of notional; a new tranche is opened every day and held for the life of the long repo. Net 4 bp: after a bid–ask spread of 4 basis points in rate terms on each change of a tranche, the value used by DST. Net obs.: after the observed spread of the pair, the average of the mean bid–ask spreads of its two repo maturities over the sample, computed by the author from Bloomberg bid and ask quotes (column Spread, in basis points). Best: the largest fee among the 22 forecasts in each column and pair (chosen ex post; the best forecast differs between columns). Break-even: the bid–ask spread at which the fee falls to zero (0 if it is not positive before costs). Perfect foresight: $\lvert xr - \overline{xr}\rvert$ is the average absolute deviation of the realised excess return from the EH forecast (prevailing mean) $\overline{xr}$, in basis points a year; it is not a bound on the fee, because positions are scaled by $\bar{\sigma}_t/\widehat{\sigma}_t$. Gross, Net, and Break-even refer to an investor with perfect foresight of the sign of the excess return in the same strategy. The gross fee bounds the gross fee of any forecast in this strategy. The net fees are not bounds: perfect foresight changes position whenever the sign changes and pays more spread than a forecast that rarely trades. Out-of-sample forecasts from five lags of the two rates of each pair.

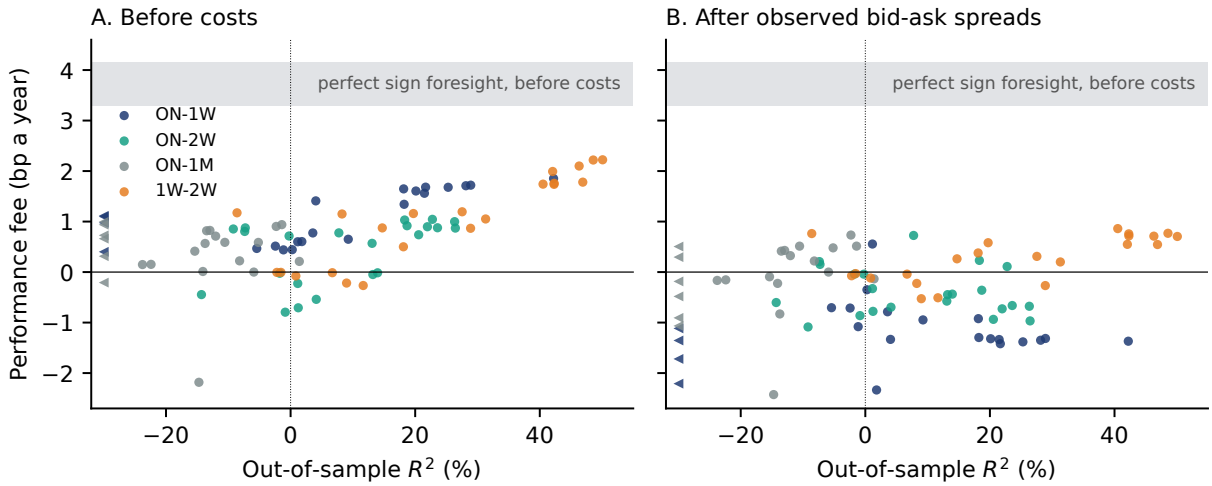


Figure 1: **Statistical Predictability versus Economic Value.** Each point is one of the 22 forecasts (the forecast from a vector autoregression, VAR, 19 machine-learning models, the equal-weighted combination of the VAR and the 18 individual machine-learning models, and an exponentially weighted mean of past excess returns) for one pair of repo maturities. Horizontal axis: out-of-sample R^2 relative to the forecast under the expectations hypothesis (EH), the prevailing mean (Supplement, Table S6); values below -30% are shown at -30% as triangles. Vertical axis: annual performance fee in basis points per unit of notional that an investor with relative risk aversion of 5 would pay to switch from the EH strategy to the forecast-based strategy (Table 2), before costs (Panel A) and after the observed bid–ask spread of each pair (Panel B). The grey band spans the fee of an investor with perfect foresight of the sign of the excess return, before costs, across the four pairs, which bounds the gross fee of any forecast in this strategy.

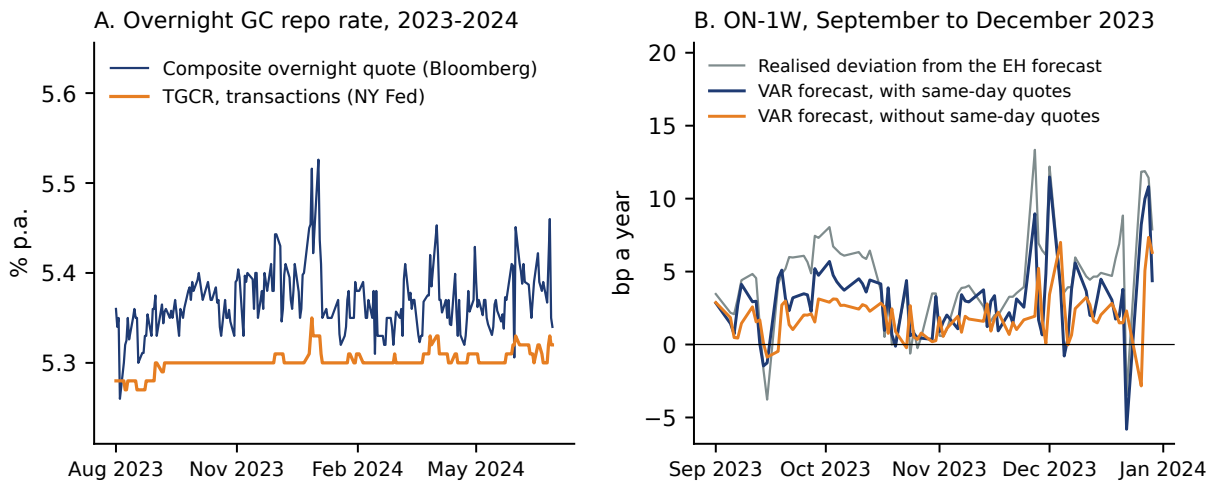


Figure 2: **Composite Quotes, Transaction Rates, and Forecasts.** Panel A: Bloomberg composite closing quote of the overnight general-collateral (GC) repo rate and the tri-party general collateral rate (TGCR), a volume-weighted median of overnight tri-party repo transactions published by the Federal Reserve Bank of New York, in percent per annum, August 2023 to June 2024, a period in which the target range of the federal funds rate was unchanged; since April 2018 the composite changes on 89% of business days and the TGCR on 49% (Supplement, Table S11). Panel B: for the overnight versus one-week pair (ON-1W), the realised excess return and the forecasts of the VAR with the same-day rates among the predictors (lags t to $t-4$) and without them (lags $t-1$ to $t-5$), each minus the EH forecast, in basis points a year, September to December 2023 (Table 1, Panel C).

The Limits of the Expectations Hypothesis

Supplementary Material

This supplement documents the forecasting protocol and the evidence behind the three results of the letter. All tables come from the same programs, data, and samples as the letter. The research uses the overnight, one-week, two-week, and one-month repo rates (pairs ON-1W, ON-2W, ON-1M, and 1W-2W), the four Bloomberg composites that are still updated in 2026; the 3W, 2M, and 3M composites have not changed since December 2015. The stale-quote filter for the four maturities is unaffected by the exclusion of the other three. Let $i_t^{(n)}$ be the simple ACT/360 rate at date t on a repo with a maturity of n business days, and let the short repo have maturity m, with $n = km$ for an integer k. Roll $j = 0, \ldots, k-1$ starts on the business day $t_j = t + jm$ and lasts d_j calendar days, and the long repo lasts $D_t = \sum_j d_j$ calendar days. The annualised excess return of rolling the short repo over holding the long repo is

$$xr_t^{(n,m)} = \left(r_t - r_t^f\right) \frac{360}{D_t}, \qquad r_t^f = i_t^{(n)} \frac{D_t}{360}, \qquad r_t = \prod_{j=0}^{k-1} \left(1 + i_{t_j}^{(m)} \frac{d_j}{360}\right) - 1, \tag{S1}$$

and the expectations hypothesis (EH) with a constant term premium $\pi^{(n,m)}$ implies that $E_t[xr_t^{(n,m)}]$ equals $-\pi^{(n,m)}$, so that the prevailing mean is the EH forecast. Section S2 refers to the vector autoregression (VAR) restrictions of Della Corte, Sarno, and Thornton (2008) (DST),

$$e_2' = e_1' \frac{1}{k} \sum_{j=0}^{k-1} G^{mj}, \tag{S2}$$

where G is the companion matrix of a bivariate VAR(5) of the short and the long rate and e_1, e_2 select the two rates.

S1 Forecasting Protocol and Models

Figure S1 shows the repo quotes used in the analysis. Table S1 lists the elements of the forecasting design, which are identical for every model. Table S2 summarises the model classes and the departures from the EH they can represent, and Table S3 the hyperparameter grids. Abbreviations in the tables: OLS, ordinary least squares; PCR and PLS, principal-component and partial least squares regression; SVR, support vector regression with a linear or radial basis function (RBF) kernel; KNN, nearest neighbours; NN (8-4-2), a neural network with hidden layers of 8, 4, and 2 units; EWMA, exponentially weighted mean of past excess returns; SPA, test of superior predictive ability; MCS, model confidence set; DM, Diebold–Mariano test.

The forecasting protocol step by step

Targets and timing. The target of the pair with short maturity m and long maturity n is the annualised excess return $xr_t^{(n,m)}$ of Equation (S1). It is indexed by its origin t and realised n business days later; the horizon n is 5 (ON-1W), 10 (ON-2W, 1W-2W), and 21 (ON-1M). US bond-market holidays are removed from the calendar, so all steps, lags, and windows count business days. The target is missing if any quote that enters it is missing. The predictors at t are the two rates of the pair at lags 0 to 4, that is, quotes dated t to $t-4$.

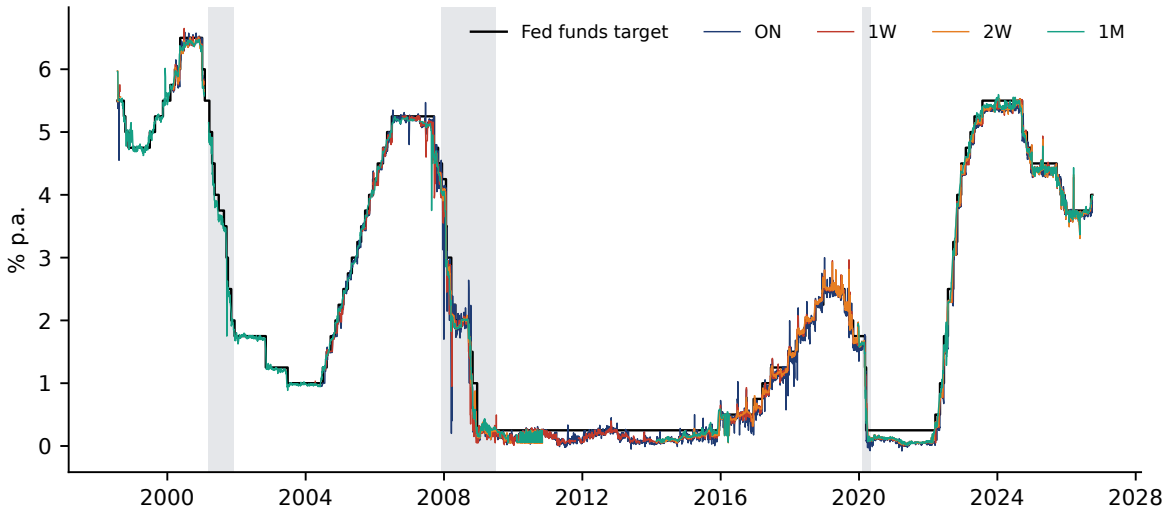


Figure S1: **Repo Quotes and the Federal Funds Target.** Daily composite quotes of the overnight, one-week, two-week, and one-month general-collateral repo rates used in the analysis and the federal funds target (upper bound of the range from December 2008), in percent per annum, 23 July 1998 to 30 September 2026. Gaps are stale quotes, runs of identical values longer than 20 business days, which are removed. Shaded areas: recessions dated by the National Bureau of Economic Research (NBER).

Table S1: **Common Forecasting Protocol**

Element	Specification (all models)
Target	Annualised excess return of rolling the short over holding the long repo, Equation (S1)
Benchmark	EH forecast: prevailing mean of the training targets at the last re-estimation
Predictors	Five daily lags of the two rates of the pair (DST's VAR(5) information set); same for every model
Forecast origin	Every business day t that is not a US holiday; information dated t or earlier
Estimation window	Rolling 1,250 business days; only pairs (z_s, xr_s) of predictors and target with $s+n \leq t$, so no target is realised after t; at least 500 usable observations
Re-estimation	Every 21 forecast dates; hyperparameters re-tuned every 252 forecast dates
Tuning	Last 15% of the estimation window, separated from the training targets by $n-1$ observations; objective: mean squared error; grids in Table S3
Preprocessing	Standardisation with the mean and standard deviation of the estimation window; standardised predictors clipped at ± 5; principal components (PCR) estimated inside the window
Missing data	Repeats in runs of more than 20 identical business days (stale quotes) set to missing, and the first quote of such a run as well if it differs by more than 0.5 percentage points from the median of the other maturities; isolated reverting spikes of more than 2 percentage points (typos) set to missing; no imputation; a forecast requires all predictors
Stochastic training	Each network forecast averages three initialisations (one in tuning); the ensemble of five architectures and the equal-weighted combination of the 19 individual forecasts (VAR, 13 methods, five networks) are fixed ex ante
Evaluation	R^2_{oos} on common dates; Clark–West, Diebold–Mariano, and encompassing tests with Newey–West variances (automatic bandwidth, at least n lags); SPA and model confidence set over all forecasts; Holm adjustment of the Clark–West and encompassing tests

Tuning budgets differ across classes, from no tuning (VAR, bagging) to 40 configurations for the networks (eight per architecture); Table S3 lists the grids. The forecasting design was fixed on data to April 2021 and applied unchanged to the holdout from 21 April 2021; the choice of maturities and the treatment of holidays were revised in 2026 on data-quality grounds.

Estimation window. At origin t, a model may use the pair (z_s, xr_s) of predictors and target only if $s+n \leq t$. The last usable target is therefore realised by t, and the most recent $n-1$ origins before t are not used. The window covers the 1,250 origins $s = t-n-1249, \ldots, t-n$. Only origins with complete predictors and a non-missing target enter the estimation. A model is estimated only if at least 500 such observations remain.

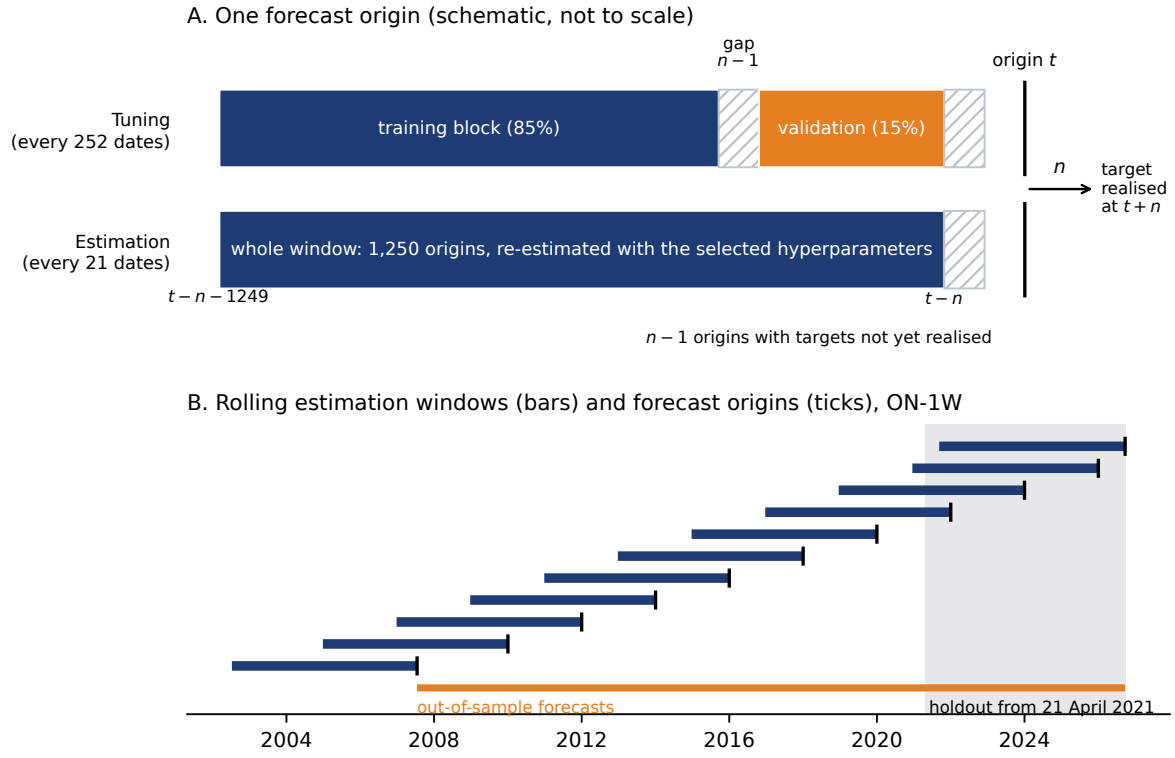


Figure S2: **Sample Splitting.** Panel A: the data used for a forecast at origin t of an excess return realised n business days later. The estimation window contains the 1,250 origins $t-n-1249$ to $t-n$, whose targets are realised by t; the $n-1$ most recent origins are not used. Every 252 forecast dates, hyperparameters are tuned on the last 15% of the window, separated from the training block by $n-1$ origins so that no training target overlaps a validation target; every 21 forecast dates, the model is re-estimated on the whole window with the selected hyperparameters. Scaling, principal components, and early stopping use the same blocks. Panel B: estimation windows (bars) and forecast origins (ticks) of the ON-1W pair at intervals of two years, the out-of-sample period, and the holdout from 21 April 2021, after the design was fixed on earlier data.

Table S2: **Model Classes and the Departures from the EH They Can Detect**

Class	Methods	Departures they can detect	Main limitation
EH forecast	Prevailing mean	None (constant premium)	–
VAR benchmark	OLS on lags of the two rates	Linear functions of recent rates, as in DST	Unstable with many correlated regressors
Regularised linear	Ridge, lasso, elastic net, PCR, PLS, linear SVR	Linear functions; robust to collinearity and noise	No nonlinearities or interactions
Kernel and local	RBF SVR, nearest neighbours	Smooth nonlinear relations, local patterns	Curse of dimensionality
Tree-based	Tree, random forest, boosting, bagging	Thresholds, regimes, interactions	Piecewise constant, poor extrapolation
Neural networks	1 to 5 hidden layers, ensemble	Smooth nonlinearities and interactions of any form	Need large signal-to-noise and regularisation
Reference forecasts	Combination of the 19 individual forecasts; EWMA of past excess returns	Robust aggregation; slowly moving premium	No new information

All models use the same predictors, estimation windows, re-estimation dates, and hyperparameter tuning on the same validation block (Section S1). Under the EH, none of them can forecast the excess return of rolling over holding better than the prevailing mean. Hyperparameter grids are in Table S3.

Re-estimation and tuning. Models are re-estimated every 21 forecast origins. Between re-estimations, the latest fitted model is applied to the new predictors. Hyperparameters are re-tuned every 252 origins, about once a year. Tuning splits the estimation sample chronologically. The last 15% of the observations form the validation block. The $n-1$ observations before it

Table S3: **Models and Hyperparameters**

Model	Hyperparameters
OLS / VAR	not tuned
Ridge	`alpha` $\in \{10^{-2}, 10^{-1.5}, \ldots, 10^{3}\}$ (11 points)
Lasso	`alpha` $\in \{10^{-4}, 10^{-3.5}, \ldots, 10^{0}\}$ (9 points)
Elastic net	`alpha` $\in \{10^{-4}, 10^{-3.5}, \ldots, 10^{0}\}$ (9 points); `l1_ratio` $\in$ {0.1, 0.5, 0.9}
PCR	`n_components` $\in$ {1, 2, 3, 5, 8}
PLS	`n_components` $\in$ {1, 2, 3, 5}
Linear SVR	`C` $\in \{10^{-3}, 10^{-2}, \ldots, 10^{1}\}$ (5 points); `epsilon` $\in$ {0, 0.1}
RBF SVR	`C` $\in$ {0.1, 1, 10}; `gamma` $\in$ {scale, 0.01}; `epsilon` $\in$ {0.05, 0.2}
KNN	`n_neighbors` $\in$ {5, 10, 20}; `weights` $\in$ {uniform, distance}
Tree	`max_depth` $\in$ {2, 3, 5}; `min_samples_leaf` $\in$ {5}
Random forest	`max_depth` $\in$ {3, 6, ∞}; `max_features` $\in$ {0.33, 1}; `min_samples_leaf` $\in$ {5}; fixed: `n_estimators`=300
Boosting	`n_estimators` $\in$ {100, 300}; `learning_rate` $\in$ {0.01, 0.1}; `max_depth` $\in$ {1, 2, 3}; fixed: `subsample`=0.8
Bagging	not tuned; fixed: `n_estimators`=100
Random features (ridge)	`n_features` $\in$ {100, 1000, 5000}; `gamma` $\in$ {0.02, 0.1}; `alpha` $\in$ {0.1, 10, 1000}
Neural networks	architectures (2), (4-2), (8-4-2), (16-8-4-2), (32-16-8-4-2); `dropout` $\in$ {0, 0.2}; `l1l2` $\in$ {0.0001, 0.001}; `learning_rate` $\in$ {0.001, 0.01}; Adam, batch 256, early stopping with patience 10, at most 200 epochs, 3 seeds

Forecasting models and hyperparameter grids (daily data). Each model is estimated on predictors and a target standardised with the mean and standard deviation of the estimation window; standardised predictors are winsorised at ± 5. Hyperparameters are chosen on the last 15% of the estimation window by minimising the mean squared forecast error. Each hidden layer of a network applies a dense layer with an elastic-net penalty, batch normalisation, a ReLU activation and dropout; forecasts average the networks of the different seeds. The table is generated from the code.

are dropped, so no training target overlaps a validation target. Each grid point is estimated on the remaining observations. The point with the lowest mean squared error in the validation block is selected, and the model is then re-estimated on the whole window.

Preprocessing. Every model is estimated on standardised predictors and a standardised target. Means and standard deviations come from the estimation sample only, and in tuning from the training block only. Standardised predictors are clipped at ± 5. Forecasts are transformed back to the units of the target.

Benchmark, missing data, and storage. The EH forecast is the mean of the training targets at the last re-estimation. Missing quotes are not imputed. An origin with a missing predictor yields no forecast. If fewer than 500 usable observations remain at a scheduled re-estimation, the previous model and EH forecast are kept. For every pair, model, and origin, the code stores the realised target, the EH forecast, and the model forecast, together with the selected hyperparameters. All forecasts are evaluated on the origins common to all models with a realised target.

Look-ahead. Figure S2 shows how each window is split. Every quantity that a forecast depends on is estimated from information available at its origin: the training pairs, the means and standard deviations of the scaling, principal components, hyperparameters, early stopping, the EH forecast, and the EWMA. The same holds for expanding windows, which change only the start of the window. Three choices use later data. Stale quotes are classified over the whole sample, which determines the estimation and evaluation dates but not how a forecast is computed; a rule that uses only past quotes gives similar results (Table S18). The other two concern only the forecasts with macro-financial predictors (Table S20, Panel B, and Figure S7, Panel A): the extra difference of nine of these series was chosen with a stationarity test on the whole sample (Section S8), and the macroeconomic series are revised data, entered with a publication lag of three months rather than as real-time vintages; the financial series are

observed at the end of each month. Monthly values are used only from their publication date onward.

Neural networks and the other models

Architectures. We estimate five feed-forward networks with hidden layers of (2), (4-2), (8-4-2), (16-8-4-2), and (32-16-8-4-2) units (Gu, Kelly, and Xiu, 2020). Each hidden layer applies a dense layer, batch normalisation (Ioffe and Szegedy, 2015), a ReLU activation, and dropout. The output layer is a single linear unit.

Training. The loss is the mean squared error plus equal L1 and L2 penalties on the weights of the hidden layers. The optimiser is Adam (Kingma and Ba, 2015) with mini-batches of 256 and at most 200 epochs. Training stops early if the validation loss does not improve for 10 epochs, and the best weights are restored. The early-stopping block is the last 15% of the training sample, separated by $n - 1$ observations. The networks are thus fitted on about 85% of each window. The grid comprises dropout $\in \{0, 0.2\}$, penalty $\in \{10^{-4}, 10^{-3}\}$, and learning rate $\in \{10^{-3}, 10^{-2}\}$.

Seeds and ensemble. Each network forecast averages three independently initialised networks. They are trained jointly as separate towers and share only the early-stopping epoch. In tuning, a single network is used per grid point. The NN ensemble is the equal-weighted mean of the five architectures.

Other models. The VAR forecast regresses the excess return directly on the ten lagged rates by OLS. Regularised linear models are ridge, lasso, elastic net, PCR, PLS, and linear SVR. The RBF SVR and nearest neighbours capture smooth and local nonlinearity. Tree-based models are a single tree, a random forest of 300 trees, gradient boosting with subsampling of 0.8, and bagging of 100 trees. Random features follow Kelly, Malamud, and Zhou (2024): ridge regression on 100 to 5,000 random Fourier features of the predictors. The combination is the equal-weighted mean of the forecasts of the VAR, the 13 other methods, and the five networks; the NN ensemble is not included (Rapach, Strauss, and Zhou, 2010). The EWMA is the exponentially weighted mean of all excess returns realised by t, with a half-life of 126 business days (Nagel, 2025). Until 252 returns are available, it equals the EH forecast. Table S3 lists the grids.

The horse race

Forecasts compared. The race compares 22 forecasts with the EH forecast: the VAR, 13 other methods, five networks, the NN ensemble, the combination, and the EWMA. All use the same predictors, windows, and evaluation dates.

Sequence of tests. The tests proceed from single forecasts to the whole set.

1. R^2_{oos} relative to the EH forecast (Campbell and Thompson, 2008) measures the accuracy of each forecast.
2. The one-sided test of Clark and West (2007) asks whether a forecast beats the EH forecast, which is nested in every model.
3. The SPA test of Hansen (2005) asks whether any of the 22 forecasts beats the EH forecast.
4. The 90% model confidence set of Hansen, Lunde, and Nason (2011) over all 23 forecasts asks which forecasts cannot be distinguished from the best.

5. Two-sided Diebold and Mariano (1995) tests ask whether any of the other 21 forecasts is more accurate than the VAR.

6. Encompassing tests of Harvey, Leybourne, and Newbold (1998) ask whether a learner adds information to the VAR. They cover the 19 learners, including the NN ensemble.

7. The Pesaran and Timmermann (1992) test asks whether the VAR and the NN ensemble predict the sign of the deviation of the excess return from the EH forecast. It uses non-overlapping origins.

8. The test of Giacomini and White (2006) asks whether the VAR's gain over the EH forecast depends on the state. Its instruments are a constant and a dummy for windows that contain a month turn.

9. The performance fee of Fleming, Kirby, and Ostdiek (2001), computed as in Della Corte et al. (2008), asks whether a forecast is worth money to an investor with relative risk aversion of 5.

Inference. Clark–West, Diebold–Mariano, encompassing, and Giacomini–White statistics use Bartlett (Newey–West) variances. The bandwidth follows the AR(1) rule of Andrews (1991), with at least n lags and at most a quarter of the sample. SPA and the model confidence set use 1,000 draws of the stationary bootstrap (Politis and Romano, 1994). The mean block length is 20 days or $2n$, whichever is larger.

Multiple testing. The SPA test and the model confidence set account for the search over forecasts by construction. Clark–West p-values are Holm-adjusted (Holm, 1979) within each pair (22 tests) and across all pairs (88 tests). Encompassing p-values are Holm-adjusted within each pair (19 tests). The 84 Diebold–Mariano comparisons are reported without adjustment.

S2 In-Sample Tests and Their Size

Table S4 reports a Wald test of DST's VAR restrictions for our sample; DST themselves use Lagrange multiplier and distance metric statistics with bootstrap p-values, because the Wald test has the worst small-sample properties (their footnote 7). Table S5 reports tests for neglected nonlinearity of the in-sample projection and the size of the Wald test in samples simulated from a two-factor model in which the EH holds up to a compounding term of at most about 2 basis points a year (about 1 at a rate of 5%; Section S7). In the simulated samples whose spread volatility is close to the data, the Wald test rejects in 86% to 100% of cases (Table S19, Panel E). The model is stylised: several persistence estimates lie at the upper bound of 0.9999, and for some pairs the simulated rates range more widely than the data. The size result should therefore be read as evidence that the asymptotic critical values are unreliable for near-integrated daily rates with overlapping returns, not as an exact size.

Table S4: **In-Sample VAR Tests of the Expectations Hypothesis**

$i^{(m)}$–$i^{(n)}$	k	Wald	p-value	$\rho(i^{(n)}, i^{(n)*})$	$\sigma(i^{(n)*})/\sigma(i^{(n)})$	Obs.
ON-1W	5	473.5	0.0000	0.9993	0.9993	5,429
ON-2W	10	247.2	0.0000	0.9998	0.9976	4,035
ON-1M	21	3134.2	0.0000	0.9983	1.0014	3,909
1W-2W	2	2446.5	0.0000	0.9995	0.9993	4,093

Wald test of the nonlinear cross-equation restrictions that the expectations hypothesis imposes on a bivariate VAR(5) of the short rate $i^{(m)}$ and the long rate $i^{(n)}$, $e_2' = e_1' k^{-1} \sum_{j=0}^{k-1} G^{mj}$ (see Equation (S2)). The covariance matrix of the VAR coefficients is heteroskedasticity robust; the statistic is $\chi^2(10)$ under the null. $i^{(n)*}$ is the theoretical long rate implied by the EH and the VAR; the last columns report its correlation with and relative volatility to the actual long rate. Daily data, stale quotes removed.

Table S5: **Preliminary Evidence: Misspecification and Size of In-Sample Tests**

	ON-1W	ON-2W	ON-1M	1W-2W
Panel A: In-sample projection on the VAR information set				
R^2 (%)	47.7	25.6	11.5	46.7
Wald test, p-value	0.000	0.000	0.000	0.000
Panel B: Tests for neglected nonlinearity				
Neural-network test, median p-value	0.624	0.148	0.235	0.616
Neural-network test, Bonferroni p-value	0.701	0.010	0.691	1.000
RESET test, p-value	0.370	0.008	0.629	0.000
Panel C: Wald test in 200 samples simulated under the EH				
Persistence of the short rate φ_u	0.5435	0.9999	0.9999	0.9999
Persistence of investors' information φ_v	0.9997	0.6197	0.7333	0.8302
Rejection rate at the 5% level (%)	86.0	94.5	96.0	97.5
95th percentile of the Wald statistic	12,489	39,283	15,028	254,708
Wald statistic in the data	473.5	247.2	3134.2	2446.5
p-value simulated under the EH	0.470	0.800	0.340	0.840
Obs.	5,421	4,004	3,836	4,078

Panel A: in-sample regression of the excess return of rolling the short repo over holding the long repo on a constant and five daily lags of the two rates (the information set of the VAR of DST), and the p-value of the Wald test that all slopes are zero. Panel B: tests of the null that this linear projection is correctly specified. The neural-network test of Lee, White, and Granger (1993) regresses the residuals on the regressors and the second to fourth principal components of the activations of 20 logistic hidden units with random weights; it is repeated for ten draws of the weights, and the Bonferroni p-value is ten times the smallest p-value. The RESET test of Ramsey (1969) adds the squared and cubed fitted values. All tests use Newey–West covariances with the automatic bandwidth of Andrews (1991), at least the forecast horizon. Panel C: rates simulated from a two-factor model in which the EH holds up to a compounding term of at most about 2 basis points a year (about 1 at a rate of 5%; Section S7), $u_{t+1} = \varphi_u u_t + v_t + \varepsilon_{t+1}$, $v_{t+1} = \varphi_v v_t + \nu_{t+1}$, with u_t the deviation of the short rate from its mean and the long rate equal to the average expected short rate plus a constant; φ_u and φ_v are estimated by Gaussian maximum likelihood, the shocks are drawn by a wild bootstrap of the estimated shocks, and the simulated rates have the stale-quote pattern of the data. The Wald test of Table S4 is computed on each sample (asymptotic 5% critical value of the $\chi^2(10)$ distribution: 18.3); the simulated p-value is the share of simulated statistics at least as large as in the data. Daily data, 23 July 1998 to 30 September 2026.

S3 All Forecasts and Joint Tests

Table S6 reports the out-of-sample R^2 of all forecasts, including a pooled network estimated on the stacked observations of all pairs. Table S7 reports the tests across models (SPA, model confidence set, Holm adjustment, encompassing, direction, and conditional predictive ability). Table S8 compares each learner with the VAR, and Table S9 shows how far each learner's forecasts depart from the EH forecast relative to the VAR. Figure S3 shows when the gains accrue.

Table S6: **Out-of-Sample Predictability of Departures from the EH**

	ON-1W	ON-2W	ON-1M	1W-2W
VAR (DST)	42.21***	26.45***	−15.38	50.08***
Ridge	28.91***	23.58***	−13.48	48.58***
Lasso	25.29***	13.20***	−23.78	42.31***
Elastic net	28.14***	13.96***	−22.40	42.29***
PCR	21.48***	20.56***	−12.95	42.06***
PLS	20.12***	21.96***	−12.03	46.32***
Linear SVR	21.67***	26.33***	1.40**	46.90***
RBF SVR	18.22***	18.69***	−61.39**	27.54***
KNN	1.80***	13.08***	−13.72**	28.89***
Tree	−105.73	−7.37***	−79.51	−8.60***
Random forest	−58.62	−0.24***	−54.65*	31.31***
Boosting	−38.61*	−7.27***	−41.40*	19.71***
Bagging	−87.30	−9.19***	−42.61**	8.26***
Random features	4.04***	18.31***	−60.62	14.72***
NN (2)	9.26***	4.12***	−14.08	8.99***
NN (4-2)	−1.15	1.15**	−14.72	11.65***
NN (8-4-2)	−5.43**	−1.74**	−10.55	−2.21*
NN (16-8-4-2)	−2.44	−14.31	−8.19	−1.55
NN (32-16-8-4-2)	0.26*	−0.86*	−5.90	0.85**
NN ensemble	3.57***	1.23***	−5.17	6.70***
Combination	18.15***	22.76***	−1.41*	40.52***
EWMA mean	1.14***	7.75***	−2.31**	18.11***
Pure EH (zero)	−4.66	−11.48	−4.49	17.28***
Spread only	36.92***	26.18***	1.67***	46.74***
Pooled ridge	20.93***	28.07***	−13.83**	35.76***
Pooled NN ensemble	4.44***	4.87***	−10.63	7.57***
Obs.	4,688	3,276	3,110	3,331

R^2_{oos} (in percent) of forecasts of the annualised excess return of rolling the short repo $k = n/m$ times over holding the long repo, relative to the prevailing mean, which is the forecast implied by the expectations hypothesis with a constant term premium. Stars: test of Clark and West (2007) of H_0: equal population mean squared error, at the 10%, 5% and 1% level (Newey–West variance with the automatic bandwidth of Andrews (1991), at least the horizon). Forecasts are made every business day from rolling five-year windows; models are re-estimated every 21 business days and re-tuned every 252. Within a table all models are evaluated on the same dates. 'Pure EH (zero)' is the zero forecast, i.e. the EH with a zero term premium. Predictors: five lags of the two rates of each pair (the VAR information set). Pooled: one model estimated on the stacked, pair-wise standardised observations of all pairs with the maturities as additional inputs (Section S3); the network ensemble averages the 8-4-2 and 32-16-8-4-2 architectures; evaluated on the dates on which both the pooled and the pair forecasts exist, and not reported (–) for fewer than 200 dates. Spread only: regression on the spread between the long and the short rate at t, the predictor of the classical regression tests, evaluated on the dates on which both the spread and the pair forecasts are available.

Table S7: **Joint Tests across Models**

	ON-1W	ON-2W	ON-1M	1W-2W
Panel A: Does any model beat the EH?				
SPA p-value (best model vs EH)	0.007	0.000	0.909	0.010
MCS p-value of the EH forecast	0.412	0.153	0.968	0.143
MCS p-value of the VAR	1.000	1.000	0.889	1.000
Models in the 90% MCS	23/23	21/23	23/23	19/23
Clark–West rejections, unadjusted	15/22	20/22	5/22	20/22
Clark–West rejections, Holm within pair	14/22	18/22	0/22	19/22
Clark–West rejections, Holm across pairs	13/22	16/22	0/22	9/22
Panel B: Does machine learning add to the VAR?				
Learners not encompassed by the VAR	0/19	8/19	14/19	7/19
after Holm adjustment	0/19	3/19	0/19	0/19
Panel C: Direction and state dependence				
Hit rate of the VAR (%)	76.1***	72.3***	60.4**	71.6***
Hit rate of the NN ensemble (%)	61.4***	57.3**	47.7	56.3
Giacomini–White p-value (VAR vs EH)	0.012	0.000	0.189	0.000
Obs.	4,688	3,276	3,110	3,331

Panel A: p-value of the test of superior predictive ability of Hansen (2005) of the null that no model forecasts the excess return better than the EH forecast (prevailing mean); p-values of the EH forecast and of the VAR in the model confidence set of Hansen et al. (2011) (a forecast is in the 90% set if its p-value is at least 0.10) and the size of the set; and the number of models for which the Clark–West test rejects at the 5% level, without adjustment and with the Holm (1979) adjustment within each pair and across all pairs. The model set comprises the VAR, 19 learners (13 methods, five network architectures, and their ensemble), the equal-weighted combination of the VAR, the 13 methods, and the five networks, and an exponentially weighted mean of past excess returns; losses are squared errors, and the bootstraps use the stationary bootstrap of Politis and Romano (1994) with a mean block length of 20 days or twice the forecast horizon, whichever is larger. Panel B: number of learners (including the network ensemble) for which the encompassing test of Harvey et al. (1998) rejects the null that the VAR encompasses the learner at the 5% level. Panel C: share of non-overlapping forecasts that predict the sign of the deviation of the excess return from the EH forecast correctly, with stars for the test of Pesaran and Timmermann (1992), and the p-value of the conditional predictive ability test of Giacomini and White (2006) with a constant and a month-turn dummy as instruments. Lags of the pair as predictors.

Table S8: **Machine Learning versus the VAR Benchmark**

	ON-1W	ON-2W	ON-1M	1W-2W
Ridge	−1.24	−2.16**	0.66	−0.74
Lasso	−1.19	−3.83***	−0.93	−1.49
Elastic net	−1.22	−3.54***	−0.78	−1.50
PCR	−1.35	−3.59***	0.87	−2.10**
PLS	−1.50	−4.24***	1.12	−4.23***
Linear SVR	−1.37	−0.06	1.65*	−1.03
RBF SVR	−1.57	−2.10**	−1.49	−3.01***
KNN	−2.39**	−1.99**	0.09	−3.40***
Tree	−1.42	−2.03**	−2.17**	−3.98***
Random forest	−1.45	−2.50**	−1.69*	−2.97***
Boosting	−1.66*	−1.79*	−1.12	−2.69***
Bagging	−1.36	−1.65*	−1.19	−3.63***
Random features	−1.76*	−1.57	−1.54	−3.92***
NN (2)	−2.10**	−3.99***	0.10	−4.07***
NN (4-2)	−2.60***	−4.13***	0.05	−4.79***
NN (8-4-2)	−2.65***	−4.55***	0.33	−4.36***
NN (16-8-4-2)	−2.63***	−6.14***	0.56	−4.07***
NN (32-16-8-4-2)	−2.59***	−4.36***	0.77	−3.95***
NN ensemble	−2.35**	−4.28***	0.79	−4.03***
Combination	−1.35	−1.02	1.33	−2.29**
EWMA mean	−2.57**	−2.79***	0.75	−3.95***

Statistics of the test of Diebold and Mariano (1995) (Newey–West variance with the automatic bandwidth of Andrews (1991), at least the horizon) comparing each model with the VAR benchmark, a direct-forecast analogue of the VAR(5) of Della Corte et al. (2008) with five lags of the two repo rates. Positive values indicate a lower mean squared forecast error than the VAR. Stars: two-sided significance at the 10%, 5% and 1% level. All models use the lags of the pair as predictors.

Table S9: **How the Learners Differ from the VAR**

	ON-1W	ON-2W	ON-1M	1W-2W
Panel A: Dispersion of departures relative to the VAR				
Ridge	0.89	0.89	0.95	0.94
Lasso	0.88	0.70	0.80	0.85
Elastic net	0.89	0.71	0.80	0.84
PCR	0.88	0.90	0.97	0.91
PLS	0.85	0.90	0.98	0.97
Linear SVR	0.94	0.97	0.61	0.93
RBF SVR	0.58	0.70	1.70	1.05
KNN	0.54	1.13	1.42	0.95
Tree	1.51	1.46	1.87	1.30
Random forest	1.20	1.36	1.80	1.00
Boosting	1.10	1.37	1.83	1.16
Bagging	1.52	1.79	1.93	1.30
Random features	0.83	0.70	1.64	0.95
NN (2)	0.35	0.54	0.65	0.22
NN (4-2)	0.23	0.45	0.48	0.49
NN (8-4-2)	0.51	0.47	0.75	0.31
NN (16-8-4-2)	0.25	0.81	0.43	0.16
NN (32-16-8-4-2)	0.10	0.30	0.20	0.09
NN ensemble	0.18	0.41	0.38	0.18
Combination	0.55	0.62	0.76	0.66
EWMA mean	0.20	0.44	0.78	0.36
Panel B: Correlation of departures with the VAR				
Ridge	0.94	0.94	0.98	0.98
Lasso	0.92	0.74	0.81	0.87
Elastic net	0.94	0.75	0.83	0.86
PCR	0.87	0.89	0.97	0.91
PLS	0.87	0.92	0.97	0.98
Linear SVR	0.88	0.88	0.83	0.90
RBF SVR	0.68	0.59	0.37	0.66
KNN	0.27	0.41	0.22	0.62
Tree	0.12	0.31	0.27	0.46
Random forest	0.19	0.35	0.29	0.64
Boosting	0.23	0.30	0.21	0.57
Bagging	0.21	0.41	0.21	0.56
Random features	0.54	0.47	0.24	0.60
NN (2)	0.41	0.48	0.37	0.39
NN (4-2)	0.19	0.25	−0.05	0.37
NN (8-4-2)	0.10	0.31	0.50	0.20
NN (16-8-4-2)	0.15	0.17	0.39	0.10
NN (32-16-8-4-2)	0.13	0.15	0.40	0.20
NN ensemble	0.32	0.34	0.45	0.40
Combination	0.78	0.79	0.66	0.86
EWMA mean	0.19	0.20	0.12	0.54
Panel C: Distinct tuned configurations				
Ridge	8	5	6	6
Lasso	6	5	6	7
Elastic net	8	5	8	8
PCR	3	4	4	3
PLS	4	4	4	3
Linear SVR	7	5	5	7
RBF SVR	9	5	6	6
KNN	6	5	4	5
Tree	3	3	2	3
Random forest	5	5	5	6
Boosting	10	7	8	7
Random features	8	8	7	8
NN (2)	7	6	6	6
NN (4-2)	7	6	5	6
NN (8-4-2)	7	6	7	7
NN (16-8-4-2)	8	6	6	7
NN (32-16-8-4-2)	8	6	6	6
Number of tuning dates	19	12	9	12

Panel A: standard deviation of the departures of each model's forecast from the EH forecast (prevailing mean), relative to that of the VAR; values below one indicate forecasts that are shrunk towards the EH. Panel B: correlation of these departures with those of the VAR. Panel C: number of distinct hyperparameter configurations selected across the tuning dates (every 252 forecast origins). Lags of the pair as predictors.

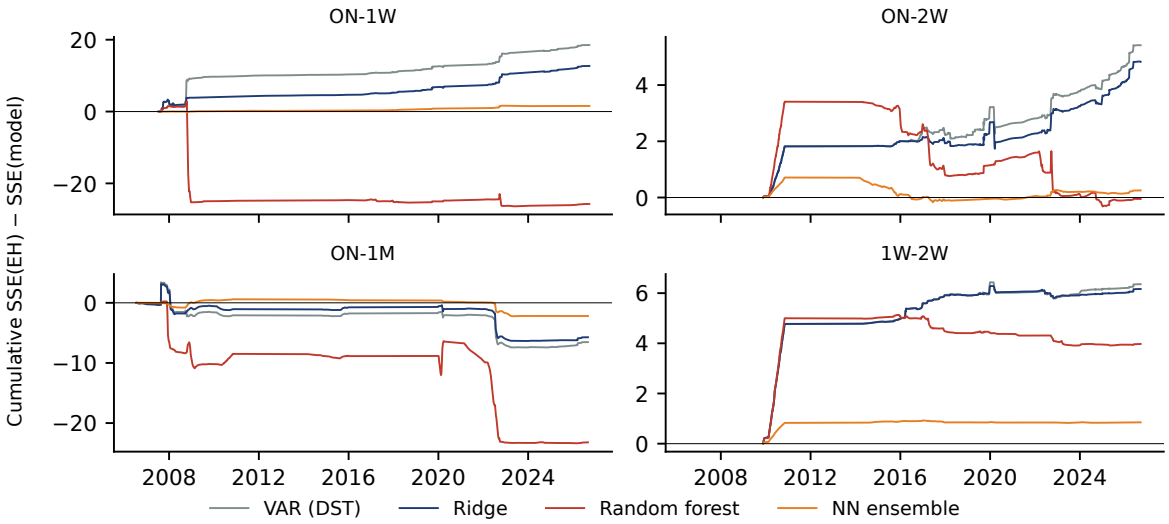


Figure S3: **Cumulative Forecast Gains Relative to the EH.** Cumulative difference between the squared forecast errors of the EH forecast (prevailing mean) and those of the VAR, ridge regression, random forest, and the neural-network ensemble. An increasing line indicates that the model forecasts better than the EH. Predictors: five lags of the two rates of each pair.

S4 Quotes, Month Turns, and a Slowly Moving Premium

Table S10 compares forecasts with and without the same-day quotes and uses an exponentially weighted mean of past excess returns as the benchmark. Table S11 compares the composite overnight quote with the transaction-based rates published by the Federal Reserve Bank of New York since April 2018. Table S12 adds calendar dummies to the VAR and to boosted trees and shows where the VAR's gains arise.

Table S10: **Same-Day Quotes and a Slowly Moving Premium**

	ON-1W	ON-2W	ON-1M	1W-2W
Panel A: R^2_{oos} (%) with same-day quotes (lags t to $t-4$) and without them (lags $t-1$ to $t-5$)				
VAR (DST), lags t to $t-4$	42.22	26.51	−15.53	49.98
VAR (DST), lags $t-1$ to $t-5$	10.09***	6.24***	−18.90	28.58***
Ridge, lags t to $t-4$	28.92	23.64	−13.63	48.47
Ridge, lags $t-1$ to $t-5$	3.36***	2.58***	−17.82	28.43***
Boosting, lags t to $t-4$	−38.61	−7.23	−41.74	19.53
Boosting, lags $t-1$ to $t-5$	−30.26	−4.26***	−36.64	0.69***
Panel B: A slowly moving premium as the benchmark				
R^2_{oos} of the EWMA vs prevailing mean (%)	1.14	7.75	−2.31	18.11
Best model, R^2_{oos} vs EWMA (%)	41.55	20.27	3.63	39.05
SPA p-value (best model vs EWMA)	0.009	0.002	0.817	0.003
Models better than EWMA (Holm)	6/21	1/21	0/21	8/21

Panel A: out-of-sample R^2 relative to the EH forecast of the VAR, ridge regression, and boosted trees with five lags of the two rates dated t to $t-4$ (as in Table S6) and dated $t-1$ to $t-5$, on the same dates; stars refer to the Clark–West test. A transitory error in the quote of the long rate at t enters both the predictors and, with the opposite sign, the excess return; forecasts without same-day quotes are free of this channel. Panel B: forecasts compared with an exponentially weighted mean of past excess returns (half-life of six months) instead of the prevailing mean: R^2_{oos} of the EWMA relative to the prevailing mean, the largest R^2_{oos} of any model relative to the EWMA, the p-value of the test of superior predictive ability of Hansen (2005) of the null that no model beats the EWMA, and the number of models that beat it in one-sided Diebold–Mariano tests at the 5% level after the Holm (1979) adjustment.

Table S11: **Composite Quotes versus Transaction-Based Overnight Rates**

	TGCR	BGCR	SOFR
Observations	2,120	2,120	2,120
Mean difference, composite minus reference (bp)	4.68	4.66	2.93
Mean absolute difference (bp)	6.03	6.01	4.49
Share of days with absolute difference above 5 bp (%)	55.9	55.6	32.7
Correlation of levels	0.999	0.999	0.999
Correlation of daily changes	0.397	0.398	0.414
Days without change, composite (%)	10.8	10.8	10.8
Days without change, reference (%)	50.8	50.3	43.6
ON-1W excess return with TGCR as rolled leg: correlation	0.847		

Comparison of the Bloomberg composite overnight general-collateral repo rate with the transaction-based reference rates of the Federal Reserve Bank of New York (tri-party general collateral rate TGCR, broad general collateral rate BGCR, and secured overnight financing rate SOFR), 2 April 2018 to 30 September 2026, excluding holidays. The last row correlates the ON-1W excess return of Equation (S1) with the one in which the rolled overnight leg uses TGCR.

Table S12: **Calendar Effects and the Departures from the EH**

	ON-1W	ON-2W	ON-1M	1W-2W
Panel A: R^2_{oos} (%)				
VAR (DST)	42.21***	26.45***	−15.38	50.08***
VAR + calendar	41.07***	22.77***	−16.91	49.47***
Calendar only	−1.44**	−0.90	0.73***	−1.03**
Boosting	−38.61*	−7.27***	−41.40*	19.71***
Boosting + calendar	−79.40*	−9.08***	−26.03*	9.74***
Panel B: Diagnostics				
DM: VAR + calendar vs VAR	−1.24	−2.10**	−1.56	−0.91
Shapley share of calendar (%)	21.37	21.65	19.37	16.18
Windows with a month turn (%)	24.04	48.14	98.39	48.00
Share of VAR gain in these windows (%)	26.36	77.55	97.08	60.08
R^2_{oos} VAR, windows with a month turn	40.76	34.95	−15.35	54.87
R^2_{oos} VAR, windows without a month turn	42.76	14.37	−16.33	44.27
Obs.	4,688	3,276	3,110	3,331

Panel A reports R^2_{oos} relative to the prevailing mean, with Clark–West significance stars, for the VAR benchmark (five lags of the two rates), the VAR augmented with calendar dummies (day of the week; last and first business day of the month, quarter and year; and whether the investment window contains a month, quarter or year turn), a regression on the calendar dummies only, and boosted trees with and without the calendar dummies. All calendar information is known at the forecast origin. Panel B reports a Diebold–Mariano test of the VAR with calendar dummies against the VAR (positive: calendar dummies improve the forecast), the share of the calendar block in the absolute Shapley contributions to the out-of-sample forecasts of the VAR with calendar dummies, the share of forecast origins whose investment window contains a month turn, the share of the VAR's out-of-sample gain over the EH forecast, $\sum_t[(xr_t - \overline{xr}_t)^2 - (xr_t - \widetilde{xr}_t)^2]$, where $\overline{xr}_t$ is the EH forecast and $\widetilde{xr}_t$ the VAR forecast, earned in these windows, and the VAR's R^2_{oos} in windows with and without a month turn (in percent).

S5 Detectability of Known Departures

Table S13 and Figure S4 report how the models perform on simulated excess returns with a known linear or nonlinear departure from the EH, and Table S14 how often the Clark–West test rejects. Each cell averages three noise draws, so the rejection rates are illustrative rather than precise power estimates. The economic bound of the letter does not rely on these simulations: it is computed for a perfect-foresight investor on the actual data (Table 2 of the letter).

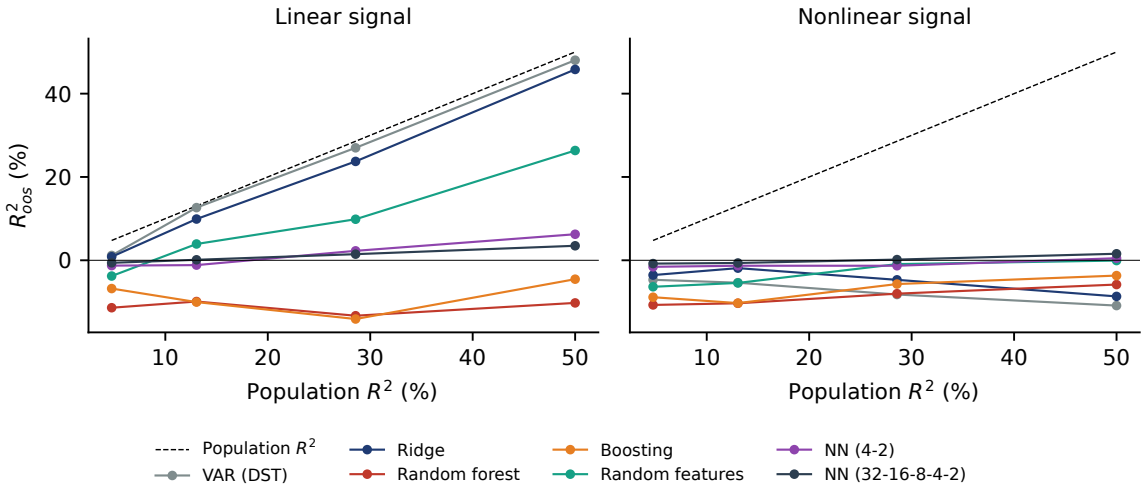


Figure S4: **Recovering a Known Departure from the EH.** Out-of-sample R^2 of the models on simulated excess returns that contain a linear (left) or nonlinear (right) departure from the EH, against the population R^2 of the departure (dashed line). Predictors, sample gaps, and backtest design of the ON-1W pair; averages over three noise draws.

Table S13: **Can the Learners Recover a Known Signal? Signal-to-Noise Simulation**

SNR	Population R^2	Oracle	VAR (DST)	Ridge	Random forest	Boosting	Random features	NN (4-2)	NN (32-16-8-4-2)
Panel A: linear signal									
0.05	4.8	5.2	1.2 (0.5)	0.8 (0.4)	−11.4 (2.3)	−6.8 (0.5)	−3.8 (2.3)	−1.3 (0.8)	−0.6 (0.2)
0.15	13.0	15.6	12.6 (0.4)	9.9 (4.6)	−9.9 (5.6)	−10.0 (7.2)	3.9 (0.8)	−1.2 (0.8)	0.1 (0.2)
0.40	28.6	29.8	27.0 (1.4)	23.7 (3.9)	−13.3 (1.6)	−14.1 (7.0)	9.9 (0.9)	2.3 (0.6)	1.5 (0.7)
1.00	50.0	49.8	48.0 (0.8)	45.8 (0.8)	−10.2 (3.0)	−4.5 (5.3)	26.4 (2.1)	6.3 (2.2)	3.5 (0.2)
Panel B: nonlinear signal									
0.05	4.8	4.9	−4.7 (0.8)	−3.5 (0.1)	−10.7 (1.8)	−8.9 (1.3)	−6.4 (2.7)	−1.6 (0.5)	−0.8 (0.2)
0.15	13.0	14.2	−5.4 (0.6)	−1.9 (0.6)	−10.3 (1.8)	−10.3 (3.4)	−5.4 (2.2)	−1.3 (1.0)	−0.6 (0.5)
0.40	28.6	31.0	−8.2 (0.3)	−4.7 (3.3)	−8.0 (0.9)	−5.7 (2.4)	−0.9 (0.8)	−1.3 (0.3)	0.2 (0.8)
1.00	50.0	51.1	−10.9 (0.9)	−8.7 (0.9)	−5.8 (3.7)	−3.7 (6.7)	−0.1 (2.6)	0.5 (0.7)	1.6 (0.3)
Panel C: linear signal, performance fee before costs (bp a year)									
0.05		0.55	0.17	0.10	−0.40	−0.24	−0.03	−0.22	−0.15
0.15		1.24	1.01	0.85	0.08	0.12	0.60	−0.09	0.14
0.40		1.32	1.00	0.73	−0.29	−0.18	0.40	−0.38	−0.15
1.00		2.04	1.95	1.85	0.00	0.31	1.19	−0.07	0.01
Panel D: nonlinear signal, performance fee before costs (bp a year)									
0.05		0.37	−0.56	−0.42	−0.70	−0.54	−0.40	−0.52	−0.23
0.15		0.82	−0.41	−0.16	−0.41	−0.29	−0.25	−0.54	−0.18
0.40		1.59	−0.64	−0.41	−0.71	−0.38	−0.18	−0.73	−0.26
1.00		2.77	−0.21	−0.06	−0.13	−0.02	0.31	−0.11	−0.10

Monte Carlo evaluation of the learners on simulated excess returns. The predictors (five lags of the ON and 1W rates), the sample with its gaps and the backtest design are those of the ON-1W pair; the target is $y_t = \sqrt{\text{SNR}}\, f(z_t) + \epsilon_t$ with $\text{Var}(f) = \text{Var}(\epsilon) = 1$, so that the population R^2 of the true conditional mean is $\text{SNR}/(1+\text{SNR})$. Panel A: linear signal, the projection of the actual ON-1W excess return on the predictors. Panel B: purely nonlinear signal, the interaction of the 1W–ON spread with the change in the ON rate, orthogonalised on the predictors. The noise ϵ_t is a moving average of five i.i.d. normal shocks, which reproduces the overlap of the excess returns. 'Oracle' is the R^2_{oos} of the true conditional mean. Entries are R^2_{oos} in percent relative to the prevailing mean, averaged over independent noise draws (standard deviation across draws in parentheses). Panels C and D: the simulated target is mapped into an excess return with the mean and volatility of the actual ON-1W excess return, and the entries are the performance fees of Table S15 before transaction costs of the oracle and of the learners, in basis points a year; with a bid–ask spread of 4 basis points, all fees, including those of the oracle, are negative.

Table S14: **Signal-to-Noise Simulation: Rejection Rates**

SNR	VAR (DST)	Ridge	Random forest	Boosting	Random features	NN (4-2)	NN (32-16-8-4-2)
Panel A: linear signal							
0.05	100	100	0	0	33	0	0
0.15	100	100	33	67	100	0	0
0.40	100	100	0	33	100	33	67
1.00	100	100	100	100	100	67	100
Panel B: nonlinear signal							
0.05	0	0	0	0	0	0	0
0.15	0	0	0	0	0	0	0
0.40	0	0	0	67	100	0	33
1.00	0	0	100	100	100	100	100

Share of the simulated samples (in percent) in which the Clark–West test rejects H_0 of equal population mean squared error at the 5% level, for the designs of Table S13.

S6 Economic Value and Robustness

Figure S5 summarises the statistical and economic results of the letter, and Figure S6 shows how the excess returns of the strategies accumulate over time. Table S15 reports the performance fees and risk-adjusted abnormal returns of DST's four cases (with and without costs and position limits). Table S16 varies risk aversion. Table S17 collects the robustness checks: episodes and loss function, out-of-sample tests under the simulated EH, stability over time (Figure S8), design choices (Table S18), the holdout, and break-even spreads. The gross fee of perfect sign foresight bounds the gross fee of any forecast in this strategy: every strategy holds a position of the same

size $|w_t| = \bar{\sigma}_t/\widehat{\sigma}_t$ and differs only in its sign, so on each date perfect foresight attains the largest return $w_t\, xr_t^{(n,m)}$, and with relative risk aversion of 5 quadratic utility increases in the portfolio return over the range of repo returns. Table S19 checks the statistical inference for the VAR: by regime, on non-overlapping origins, with bootstrap intervals for R^2_{oos}, with a model confidence set of five standard forecasts chosen after the results for the full set were known, and in the simulated samples whose spread volatility is close to the data. Panel G shows that the economic bound stays at a few basis points a year in both policy regimes: at the zero lower bound and away from it, even perfect foresight of the sign earns between 2.5 and 4.8 basis points a year before costs, and the VAR at most 3.3.

Table S20 and Figure S7 ask whether the conclusions depend on DST's information set. With the repo rates in levels, trees, nearest neighbours, and networks cannot extrapolate beyond the rates of their training window. With the same information in stationary form, the term spread and the change of the short rate, the flexible models improve markedly but, among the three predictable pairs, exceed the VAR only for ON-2W (random forest 34.1% against 26.4%); the best forecasts for ON-1W and 1W-2W are linear (OLS and linear SVR). No forecast beats the VAR on DST's set after accounting for the search (SPA p-values between 0.052 and 1), for ON-1M none beats the EH forecast (SPA $p = 0.30$), and no fee before costs exceeds 2.33 basis points a year. Adding the 100 macro-financial predictors of Section S8, similar to the panel of Ludvigson and Ng (2009), lowers the accuracy of almost every forecast, and adding one group at a time lowers the R^2_{oos} of ridge in 46 of 48 cases (Figure S7, Panel A); the larger of the two gains, output and income for ON-1W, raises it from 28.9% to 35.4%, below the VAR's 42.2%. Within DST's set, the Shapley decomposition (Panel B) gives the same-day quotes 33.5 of 42.2 percentage points of the VAR's R^2_{oos} for ON-1W, 22.2 of 26.4 for ON-2W, and 35.9 of 50.1 for 1W-2W. Because it splits the information that the two blocks share, it is not a measure of quote noise and gives the same-day quotes a similar share for all three pairs. The letter uses the loss of R^2_{oos} when the same-day quotes are dropped (Table 1 of the letter, Panel C). What distinguishes 1W-2W is what the lagged quotes predict on their own, 28.6% against 10.1% and 6.2%, mostly at the zero lower bound (Table S19, Panel A).

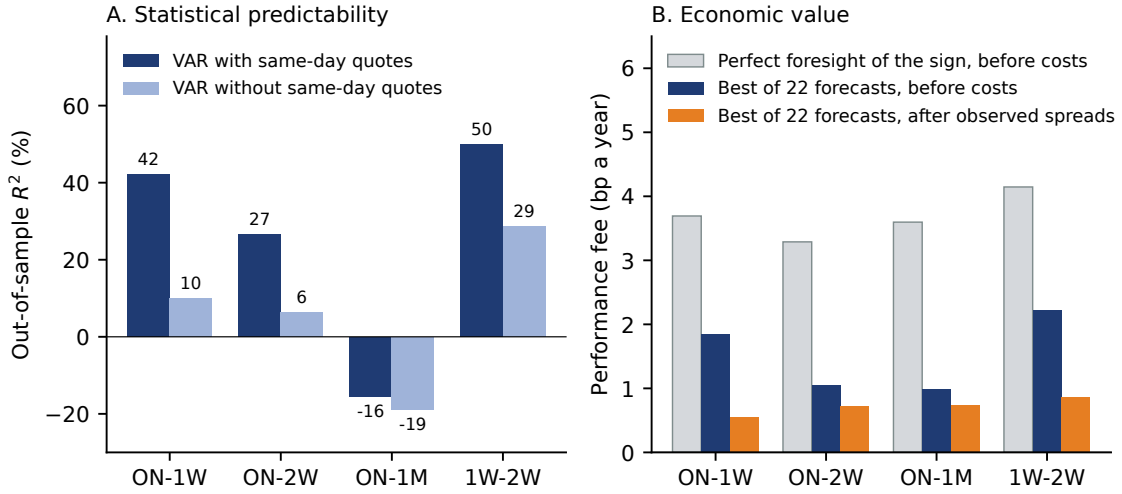


Figure S5: **Summary of the Statistical and Economic Results.** Panel A: out-of-sample R^2 of the VAR forecast relative to the EH forecast, with the same-day rates among the predictors (lags t to $t-4$) and without them (lags $t-1$ to $t-5$), on the same dates (Table 1 of the letter, Panel C). Panel B: annual performance fee in basis points per unit of notional of an investor with perfect foresight of the sign of the excess return before costs, and of the best of the 22 forecasts before costs and after the observed bid–ask spreads (Table 2 of the letter).

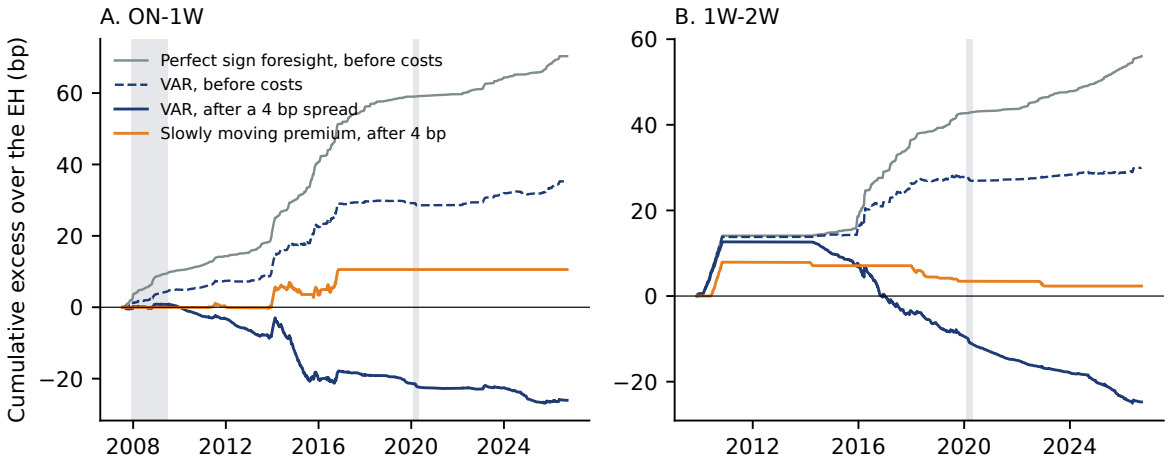


Figure S6: **Cumulative Excess Returns over the EH Strategy.** Cumulative sum of the daily excess returns of the forecast-based strategies over the EH strategy, in basis points per unit of notional, for ON-1W (Panel A) and 1W-2W (Panel B): an investor with perfect foresight of the sign of the excess return before costs, the VAR before costs and after a bid–ask spread of 4 basis points, and the exponentially weighted mean of past excess returns after 4 basis points. Positions, tranches, and costs as in Table S15; each daily tranche carries $1/n$ of the notional, so the average slope per year equals the annual performance fee before the risk adjustment. Shaded areas: NBER recessions.

Table S15: **Economic Value of Departures from the EH**

	VAR (DST)		Ridge		Random forest		Boosting		NN ensemble		Combination		EWMA mean	
	$\mathcal{F}$	M^2	$\mathcal{F}$	M^2	$\mathcal{F}$	M^2	$\mathcal{F}$	M^2	$\mathcal{F}$	M^2	$\mathcal{F}$	M^2	$\mathcal{F}$	M^2
Panel A: Case 1: no transaction costs, unconstrained weights														
ON-1W	1.85	2.06	1.72	1.90	0.98	1.04	1.10	1.17	0.77	0.82	1.65	1.81	0.60	0.63
ON-2W	0.87	1.01	0.88	1.02	0.71	0.82	0.87	1.01	−0.71	−0.75	1.04	1.22	0.78	0.89
ON-1M	0.41	0.45	0.82	0.90	0.99	1.10	0.94	1.03	0.59	0.64	0.94	1.04	0.90	0.99
1W-2W	2.22	2.34	2.22	2.34	1.05	1.07	1.16	1.18	−0.01	−0.01	1.74	1.80	0.50	0.50
Panel B: Case 2: transaction costs, unconstrained weights														
ON-1W	−1.37	−1.37	−1.31	−1.33	−1.72	−1.66	−1.35	−1.36	−0.79	−0.87	−0.92	−1.00	0.56	0.57
ON-2W	−1.23	−1.36	−0.88	−1.01	−0.15	−0.19	0.05	0.02	−0.79	−0.86	−0.02	−0.03	0.72	0.81
ON-1M	−0.17	−0.20	0.36	0.37	0.44	0.46	−0.35	−0.43	0.46	0.51	0.45	0.49	0.71	0.73
1W-2W	−1.83	−1.63	−1.65	−1.48	−1.22	−1.15	−0.38	−0.38	−0.09	−0.09	−0.61	−0.60	0.17	0.16
Panel C: Case 3: transaction costs, $-1 \leq w_t \leq 2$														
ON-1W	−1.00	−1.02	−0.96	−0.97	−1.22	−1.23	−1.01	−1.03	−0.43	−0.44	−0.58	−0.59	0.34	0.47
ON-2W	−0.98	−1.15	−0.69	−0.85	−0.03	−0.10	0.09	0.04	−0.58	−0.74	0.07	0.02	0.71	0.80
ON-1M	−0.15	−0.19	0.36	0.36	0.46	0.48	−0.18	−0.23	0.46	0.50	0.47	0.50	0.72	0.73
1W-2W	−1.34	−1.04	−1.17	−0.91	−0.89	−0.75	−0.13	−0.14	−0.08	−0.08	−0.47	−0.42	0.21	0.19
Panel D: Case 4: transaction costs, no short selling ($0 \leq w_t \leq 1$)														
ON-1W	−0.45	−0.45	−0.43	−0.43	−0.54	−0.53	−0.46	−0.45	−0.17	−0.16	−0.24	−0.24	0.12	0.14
ON-2W	−0.37	–	−0.26	–	0.04	–	0.06	–	−0.20	–	0.08	–	0.34	–
ON-1M	−0.07	−0.06	0.18	0.17	0.24	0.22	−0.03	−0.05	0.24	0.21	0.24	0.22	0.36	0.35
1W-2W	−0.53	–	−0.45	–	−0.33	–	0.02	–	−0.04	–	−0.17	–	0.12	–

A mean-variance investor with quadratic utility and relative risk aversion of 5 holds the long repo and a position $w_t = (\bar{\sigma}_t/\widehat{\sigma}_t)\,\mathrm{sign}(\widehat{\mu}_t)$ in the strategy of rolling the short repo, where $\widehat{\mu}_t$ is the forecast of the excess return, $\widehat{\sigma}_t$ its conditional volatility and $\bar{\sigma}_t$ the average conditional volatility over the estimation window, so that the median position is about one unit of notional (adapted from Della Corte et al., 2008, Section 5). A new position is opened every day and held for the life of the long repo, so that each position replaces the one opened n business days earlier; transaction costs are a bid–ask spread of 4 basis points in rate terms on this change of the position. $\mathcal{F}$ is the annualised performance fee the investor would pay to switch from the strategy implied by the expectations hypothesis (prevailing-mean forecast) to the strategy based on the model forecasts, and M^2 the risk-adjusted abnormal return of Modigliani and Modigliani (1997), both in basis points per year. Out-of-sample forecasts with the lags of the pair as predictors. M^2 is not defined (–) when the EH strategy holds no position, as in Panel D for pairs with a negative prevailing mean.

Table S16: **Economic Value: Costs, Turnover, and Risk Aversion**

	ON-1W	ON-2W	ON-1M	1W-2W
Panel A: Fee without costs (bp a year)				
VAR (DST)	1.85	0.87	0.41	2.22
Ridge	1.72	0.88	0.82	2.22
Random forest	0.98	0.71	0.99	1.05
Boosting	1.10	0.87	0.94	1.16
NN ensemble	0.77	−0.71	0.59	−0.01
Combination	1.65	1.04	0.94	1.74
EWMA mean	0.60	0.78	0.90	0.50
Panel B: Break-even bid–ask spread (bp)				
VAR (DST)	2.30	1.66	2.86	2.20
Ridge	2.27	1.99	7.09	2.29
Random forest	1.45	3.31	7.11	1.85
Boosting	1.79	4.23	2.92	3.01
NN ensemble	1.99	0.00	19.03	0.00
Combination	2.56	3.91	7.73	2.97
EWMA mean	53.04	52.87	18.53	6.14
Panel C: Change of the tranche position, $\lvert w_t - w_{t-n} \rvert$				
VAR (DST)	0.836	0.554	0.219	1.049
Ridge	0.790	0.469	0.190	1.003
Random forest	0.703	0.246	0.215	0.604
Boosting	0.642	0.236	0.393	0.424
NN ensemble	0.419	0.051	0.106	0.056
Combination	0.673	0.297	0.196	0.623
EWMA mean	0.042	0.045	0.124	0.119
Panel D: Fee with costs, risk aversion 2 (bp a year)				
VAR (DST)	−1.37	−1.23	−0.17	−1.83
Ridge	−1.31	−0.88	0.36	−1.65
Random forest	−1.72	−0.15	0.43	−1.22
Boosting	−1.35	0.05	−0.34	−0.38
NN ensemble	−0.79	−0.79	0.46	−0.09
Combination	−0.92	−0.02	0.45	−0.61
EWMA mean	0.55	0.72	0.70	0.17
Panel E: Fee with costs, risk aversion 10 (bp a year)				
VAR (DST)	−1.37	−1.23	−0.16	−1.82
Ridge	−1.32	−0.88	0.36	−1.65
Random forest	−1.72	−0.15	0.44	−1.22
Boosting	−1.36	0.05	−0.35	−0.38
NN ensemble	−0.79	−0.79	0.47	−0.09
Combination	−0.92	−0.02	0.46	−0.60
EWMA mean	0.56	0.72	0.72	0.18
Panel F: Volatility of net returns vs EH (bp a year)				
VAR (DST)	1.56	1.67	2.05	1.85
NN ensemble	1.50	1.53	1.80	0.18
Combination	1.47	1.37	2.02	1.39
EWMA mean	1.21	1.01	2.08	0.96
Panel G: Maximum drawdown vs EH (bp)				
VAR (DST)	19.94	22.77	12.01	38.25
NN ensemble	16.03	11.70	1.66	1.43
Combination	20.17	10.14	5.71	18.14
EWMA mean	4.81	0.81	4.21	6.79

Economic value of the forecasts under alternative assumptions, without position limits. The break-even spread is the bid–ask spread in rate terms at which the performance fee falls to zero. The change of the tranche position is the average absolute difference between the weight of the tranche opened at an origin and that of the tranche it replaces, opened n business days earlier, in units of notional. The baseline of Table S15 uses a spread of 4 basis points and relative risk aversion of 5. Panels F and G: returns of the strategy net of the bid–ask spread minus those of the EH strategy, measured over non-overlapping investment horizons. Lags of the pair as predictors.

Table S17: **Robustness**

	ON-1W	ON-2W	ON-1M	1W-2W
Panel A: R^2_{oos} of the VAR (%), episodes and loss function				
Full sample	42.21***	26.45***	−15.38	50.08***
Without Sep 2008–Jun 2009	37.26***	26.45***	−20.08	50.08***
Without Sep–Dec 2019	42.33***	25.54***	−15.95	51.23***
Without Mar–Jun 2020	43.55***	32.77***	−17.23	55.30***
Without year-end windows	44.71***	24.54***	−15.00	48.84***
Without all four	43.19***	32.18***	−23.57	58.04***
Largest 1% of EH errors removed	17.70***	20.78***	−34.21	46.97***
Absolute-error loss	25.33***	15.80***	−11.19	26.31***
Panel B: Simulation under the EH (200 samples)				
VAR R^2_{oos}: 95th percentile under the EH (%)	−2.56	−2.00	−6.07	−0.11
VAR R^2_{oos}: p-value simulated under the EH	0.000	0.000	0.675	0.000
Clark–West: rejection rate at 5% (%)	6.5	8.0	10.5	20.0
Panel C: Share of time with significant local gains over the EH (%)				
VAR (DST)	0.0	4.9	0.0	5.5
Combination	0.0	3.5	0.0	4.6
Panel D: Design choices (9 designs), VAR				
Lowest R^2_{oos} (%)	38.62	18.97	−15.55	39.08
Highest R^2_{oos} (%)	44.51	26.98	−0.03	52.27
Designs with Clark–West $p < 0.05$	9/9	9/9	1/9	9/9
Panel E: Holdout from 21 April 2021, R^2_{oos} (%) and performance fees				
VAR (DST)	61.28***	40.05***	−46.46	11.08***
Combination	43.14***	26.51***	−43.86	4.29***
NN ensemble	7.20*	4.26***	−23.40	0.45
Largest fee of 7 forecasts, before costs (bp a year)	1.21	0.49	0.07	0.54
Largest fee of 7 forecasts, after a 4 bp spread (bp a year)	0.06	0.00	0.00	−0.13
Panel F: Break-even bid–ask spread (bp)				
VAR (DST)	2.30	1.66	2.86	2.20
Combination	2.56	3.91	7.73	2.97

Panel A: R^2_{oos} of the VAR relative to the EH forecast after removing forecast origins whose investment window overlaps the stated periods, after removing the 1% of origins with the largest absolute errors of the EH forecast, and with absolute instead of squared errors; stars refer to the Clark–West test and, for absolute errors, to a one-sided Diebold–Mariano test. Panel B: rates are simulated from the two-factor model of Table S5, in which the EH holds up to a compounding term of at most about 2 basis points a year (about 1 at a rate of 5%; Section S7) (for the out-of-sample test with expected short rates weighted by calendar days, as in the excess return), with shocks drawn by a wild bootstrap and the stale-quote pattern of the data; on each sample the out-of-sample VAR forecasts are computed exactly as in the data. Rejection rates at the nominal 5% level measure the size of the tests; simulated p-values are the shares of simulated statistics at least as large as in the data. Panel C: share of dates at which the fluctuation test of Giacomini and Rossi (2010), computed over a rolling 30% of the out-of-sample period, indicates significantly smaller squared errors than the EH forecast at the 5% level. Panel D: range of the VAR's R^2_{oos} and number of designs with a significant Clark–West statistic across the designs in Table S18. Panel E: out-of-sample R^2 from 21 April 2021 (models, grids, windows, and tests were fixed on data until April 2021; the choice of maturities and the treatment of holidays were revised in 2026 on data-quality grounds); '–': fewer than 30 forecasts. The last two rows report the largest holdout performance fee among the forecasts of Table S15, before costs and after a bid–ask spread of 4 basis points; fees of the order of 10^{-10} are shown as 0.00. Panel F: bid–ask spread in rate terms at which the performance fee of Table S15 (no position limits) falls to zero; 0 if the fee is not positive without costs.

Table S18: **Out-of-Sample Predictability under Alternative Designs**

	ON-1W	ON-2W	ON-1M	1W-2W
VAR (DST)				
Baseline (1,250 days)	42.21***	26.45***	−15.38	50.08***
Window 750 days	40.33***	18.97***	−10.41*	48.37***
Window 2,500 days	43.43***	24.31***	−9.05*	42.93***
Expanding window	44.51***	25.30***	−0.03**	39.08***
Validation share 25%	42.21***	26.45***	−15.38	50.08***
Re-estimation every 5 days	41.71***	26.98***	−12.18	50.00***
Stale threshold 10 days	42.29***	25.72***	−9.53	47.07***
Stale threshold 40 days	38.76***	26.45***	−15.55	50.08***
Stale quotes masked in real time	38.62***	26.77***	−13.44	52.27***
Ridge				
Baseline (1,250 days)	28.91***	23.58***	−13.48	48.58***
Window 750 days	21.33***	15.95***	−14.91	45.77***
Window 2,500 days	33.97***	22.07***	−12.88	42.75***
Expanding window	35.86***	23.77***	−5.89**	40.15***
Validation share 25%	33.54***	23.80***	−15.07	48.27***
Re-estimation every 5 days	28.75***	23.19***	−10.83	48.49***
Stale threshold 10 days	32.41***	23.35***	−11.79	45.43***
Stale threshold 40 days	31.66***	23.58***	−13.46	48.58***
Stale quotes masked in real time	32.75***	26.21***	−11.23	50.84***
Lasso				
Baseline (1,250 days)	25.29***	13.20***	−23.78	42.31***
Window 750 days	26.39***	7.35***	−21.27	41.58***
Window 2,500 days	37.33***	15.57***	−5.50	42.05***
Expanding window	39.13***	19.24***	−6.82*	40.49***
Validation share 25%	24.64***	13.73***	−25.00	45.70***
Re-estimation every 5 days	25.22***	12.61***	−19.37	41.62***
Stale threshold 10 days	36.13***	20.28***	−14.98	44.92***
Stale threshold 40 days	38.75***	13.20***	−14.82	42.31***
Stale quotes masked in real time	40.26***	13.23***	−14.42	44.88***
Random forest				
Baseline (1,250 days)	−58.62	−0.24***	−54.65*	31.31***
Window 750 days	−51.93	−13.76***	−54.04**	31.23***
Window 2,500 days	−46.10	7.63***	−32.44*	−32.27***
Expanding window	−30.65*	13.21***	−23.95*	10.24***
Validation share 25%	−58.99	−1.22***	−52.85*	29.20***
Re-estimation every 5 days	−37.55*	1.18***	−47.54	32.25***
Stale threshold 10 days	−56.46	−6.22***	−55.46**	29.81***
Stale threshold 40 days	−17.86***	−0.24***	−50.23	31.31***
Stale quotes masked in real time	−14.95***	2.59***	−33.25*	34.22***
Boosting				
Baseline (1,250 days)	−38.61*	−7.27***	−41.40*	19.71***
Window 750 days	−74.04*	−11.08***	−45.64**	24.75***
Window 2,500 days	−67.94**	2.69***	−4.88**	−13.97***
Expanding window	−15.74***	17.46***	−0.15**	13.79***
Validation share 25%	−76.51	−30.34***	−56.06*	29.29***
Re-estimation every 5 days	−24.96***	3.52***	−25.99	18.83***
Stale threshold 10 days	−77.37	−32.42***	−53.80*	19.32***
Stale threshold 40 days	−106.81	−7.27***	−35.86*	19.71***
Stale quotes masked in real time	−51.43*	−12.85***	−29.89*	22.79***
Sup test across windows and validation shares, p-value				
VAR (DST)	0.000	0.000	0.082	0.000
Ridge	0.000	0.000	0.103	0.001
Lasso	0.000	0.000	0.240	0.001
Random forest	0.082	0.000	0.059	0.014
Boosting	0.025	0.002	0.034	0.010

R^2_{oos} (in percent) relative to the EH forecast under alternative designs, with Clark–West significance stars; each design is evaluated on its own out-of-sample period. The stale-quote thresholds change the evaluation sample and the network ensemble comprises the architectures estimated under each design. The sup test uses the largest Clark–West statistic across the designs that share the evaluation sample (window lengths and validation share), on their common dates, with a critical value from a joint stationary bootstrap, and so accounts for the search over designs (in the spirit of Rossi and Inoue, 2012). Lags of the pair as predictors.

Table S19: **Robustness of the Statistical Inference**

	ON-1W	ON-2W	ON-1M	1W-2W
Panel A: VAR R^2_{oos} by regime (%), with / without same-day quotes				
Zero lower bound	42.0^{***} / 7.7^{***}	38.2^{***} / 15.5^{***}	7.3^{***} / -12.1^{**}	85.1^{***} / 59.1^{***}
Away from the zero lower bound	42.3^{***} / 10.4^{***}	21.0^{***} / 1.9^{***}	-19.8 / -20.2	21.0^{***} / 3.4^{***}
Since April 2018	50.6^{***} / 10.5^{***}	26.5^{***} / 5.1^{***}	-27.2 / -30.0	7.7^{***} / -1.3^{*}
Holdout, from 21 April 2021	61.3^{***} / 15.1^{***}	40.1^{***} / 10.1^{***}	-46.5 / -56.3	11.1^{***} / -3.2^{**}
NBER recessions	31.6^{**} / 4.4^{*}	-39.5 / -21.4	-23.0 / -14.2	-102.0 / -38.4
NBER expansions	52.3^{***} / 15.4^{***}	32.9^{***} / 8.9^{***}	-7.9 / -23.7	55.2^{***} / 30.9^{***}
Panel B: VAR on non-overlapping origins only				
R^2_{oos} (%), minimum over offsets	39.4	16.2	-48.8	40.6
R^2_{oos} (%), median over offsets	41.3	26.3	-23.5	47.5
R^2_{oos} (%), maximum over offsets	46.7	37.0	16.2	62.1
Largest Clark–West p-value	0.013	0.001	0.866	0.021
Observations per offset	937	327	148	333
Panel C: 95% bootstrap interval of the VAR's R^2_{oos} (%)				
With same-day quotes	[24.1, 52.8]	[12.4, 37.6]	[−49.1, 11.3]	[29.4, 65.0]
Without same-day quotes	[−5.2, 20.8]	[−1.1, 13.3]	[−46.3, 3.3]	[12.2, 41.4]
Panel D: Model confidence set of five standard forecasts				
MCS p-value of the EH forecast	0.004	0.000	0.861	0.003
MCS p-value of the VAR	1.000	1.000	0.433	1.000
Panel E: Tests in samples simulated under the EH				
Clark–West test of the VAR, p-value simulated under the EH	0.000	0.000	0.195	0.035
Samples with spread volatility close to the data (of 200)	200	200	197	35
Wald test in these samples, rejection rate at 5% (%)	86.0	94.5	96.4	100.0
Wald test in these samples, p-value of the data	0.470	0.800	0.345	0.857
Panel F: Mean excess return of rolling over holding, bp a year (HAC t-statistic)				
All out-of-sample dates	−1.37 (−3.7)	−2.67 (−5.2)	−3.37 (−3.6)	−0.15 (−0.6)
Zero lower bound	−0.54 (−1.8)	−0.20 (−0.3)	−1.65 (−2.1)	0.88 (1.7)
Away from the zero lower bound	−2.10 (−3.5)	−4.00 (−8.1)	−4.71 (−3.4)	−0.70 (−2.9)
Since April 2018	−3.18 (−7.6)	−3.97 (−8.0)	−4.59 (−4.8)	−0.81 (−3.8)
NBER recessions	0.96 (0.4)	−14.04 (−2.9)	−11.29 (−2.9)	−3.56 (−1.7)
NBER expansions	−1.62 (−5.2)	−2.45 (−5.1)	−2.00 (−3.0)	−0.09 (−0.3)
Days in NBER recessions	457	62	457	62
Panel G: Performance fee before costs by policy regime (bp a year)				
VAR, zero lower bound	2.12	2.52	0.46	3.27
VAR, away from it	1.61	−0.02	0.38	1.67
Perfect foresight of the sign, zero lower bound	3.65	4.82	2.70	4.10
Perfect foresight of the sign, away from it	3.73	2.46	4.30	4.17

VAR: ordinary least squares on five daily lags of the pair's two rates, as in Table 1 of the letter. Panel A: R^2_{oos} relative to the EH forecast with lags t to $t-4$ (left) and $t-1$ to $t-5$ (right), on the dates of Table 1 of the letter, Panel C; stars: Clark–West test at the 10%, 5%, and 1% level. Zero lower bound: 16 December 2008 to 15 December 2015 and 16 March 2020 to 16 March 2022; away from it: all other dates. Panel B: the VAR's forecasts evaluated only on every n-th date, for each of the n possible offsets, where n is the horizon in business days, so that the excess returns do not overlap; Clark–West test with the HAC variance of Table 1 of the letter (at least one lag). Panel C: percentiles of R^2_{oos} in 1,000 stationary-bootstrap resamples of the squared forecast errors (Politis and Romano, 1994) with an average block length of $\max(20, 2n)$. Panel D: model confidence set (Hansen et al., 2011) of the EH forecast, the VAR, the term-spread regression, ridge regression, and the EWMA, with the T_{max} statistic; a forecast with a p-value below 0.10 is outside the 90% set. This set was chosen after the results for the full set (Table S7) were known, as a check on the power of the full set, which also contains forecasts with large negative R^2_{oos}. Panel E: the 200 samples of Section S2; the Clark–West p-value is the share of simulated statistics at least as large as in the data; samples with spread volatility close to the data are those within 0.75 to 1.33 times the standard deviation of the long-short spread in the data; the Clark–West p-value uses all 200 samples. Panel F: mean of the realised excess return (Equation (S1)) on the out-of-sample dates, in basis points a year; a negative mean means that holding the long repo earned more than rolling the short repo, that is, a positive term premium. NBER recessions: March to November 2001, December 2007 to June 2009, and February to April 2020 (first day of the peak month to last day of the trough month); the out-of-sample dates start after 2001, and for ON-2W and 1W-2W only the 2020 recession is covered. Panel G: annual performance fee before costs, as in Table 2 of the letter, of the VAR and of an investor with perfect foresight of the sign of the excess return, on the out-of-sample dates at the zero lower bound (as in Panel A) and on all other dates; in subperiods the perfect-foresight fee can exceed its full-sample value.

Table S20: **Extended Information Sets**

	ON-1W	ON-2W	ON-1M	1W-2W
Panel A: Stationary repo inputs (term spread and change of the short rate), R^2_{oos} (%)				
VAR (OLS)	38.3^{***}	29.9^{***}	-10.7^{*}	46.8^{***}
Ridge	32.3^{***}	26.9^{***}	-6.6^{**}	46.7^{***}
Random forest	28.5^{***}	34.1^{***}	-26.2	44.0^{***}
Boosting	26.8^{***}	31.3^{***}	-21.3	42.0^{***}
RBF SVR	25.6^{***}	33.9^{***}	-3.9^{**}	47.5^{***}
Neural-network ensemble	25.1^{***}	16.7^{***}	2.5^{**}	14.2^{***}
Combination	32.6^{***}	31.4^{***}	-0.1^{**}	44.4^{***}
Best of 19 machine-learning models	34.3^{***}	34.1^{***}	5.3^{***}	51.4^{***}
SPA p-value: no forecast beats the EH forecast	0.001	0.000	0.297	0.001
SPA p-value: no forecast beats the VAR on DST's set	1.000	0.052	0.265	0.629
Largest performance fee before costs (bp a year)	1.92	1.37	1.06	2.33
Panel B: DST's information set and 100 macro-financial predictors, R^2_{oos} (%)				
Ridge	-35.3	-33.1^{***}	-366.7	-25.5^{***}
Random forest	-50.4	7.5^{***}	-41.5	24.1^{***}
Boosting	-114.9	-37.4^{***}	-61.2	11.4^{***}
RBF SVR	-10.0	-25.4^{***}	-16.6	1.0^{***}
Neural-network ensemble	-1.8	-4.6	-5.5^{*}	2.2^{**}
Combination	-10.7	5.5^{***}	-19.9	12.1^{***}
Best of 19 machine-learning models	-1.7	7.5^{***}	-1.1^{**}	24.1^{***}
SPA p-value: no forecast beats the EH forecast	0.576	0.111	1.000	0.064
SPA p-value: no forecast beats the VAR on DST's set	1.000	1.000	0.570	1.000
Largest performance fee before costs (bp a year)	0.99	0.78	0.90	0.72
VAR on DST's information set, R^2_{oos} (%)	42.2	26.4	-15.4	50.1

All forecasts follow the protocol of Table S1 and are evaluated on the dates on which the VAR on DST's information set (five lags of the two rates) is available. Panel A: predictors are the term spread (long minus short rate) and the daily change of the short rate at lags t to $t-4$, the information of DST's set in stationary form without the level of rates. Panel B: DST's information set and the 100 macro-financial predictors of Table S21, transformed to stationarity and lagged as described in Section S8; unrestricted OLS is not estimated on 110 predictors. Stars: Clark–West test against the EH forecast at the 10%, 5%, and 1% level. SPA: tests of Hansen (2005) of the null that no forecast of the panel has a lower mean squared error than the EH forecast, and than the VAR on DST's set. Largest fee: the highest annual performance fee before costs among the forecasts of the panel (Table 2 of the letter); the perfect-foresight fee of Table 2 bounds it.

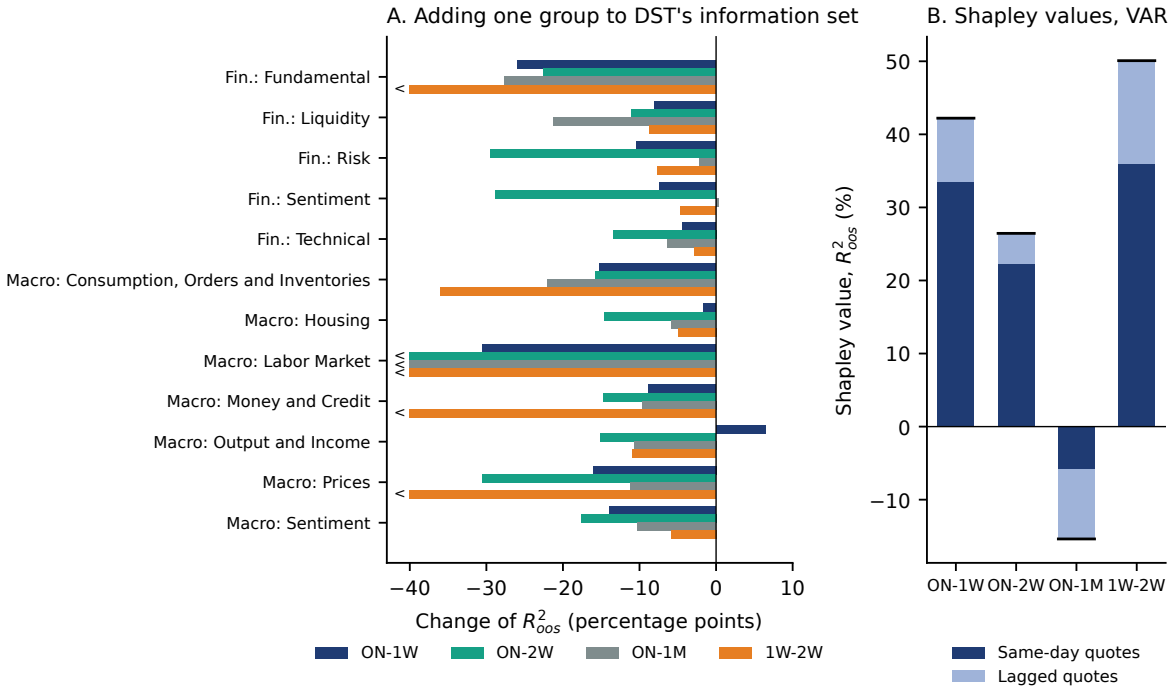


Figure S7: **Relative Importance of the Information.** Panel A: change of the out-of-sample R^2 of ridge regression, in percentage points, when one group of macro-financial predictors (Table S21) is added to DST's information set; values below -40 are shown at -40 (marked $<$). Panel B: Shapley decomposition of the out-of-sample R^2 of the VAR into the same-day quotes (lag t of both rates) and the lagged quotes (lags $t-1$ to $t-4$); each block receives the average of its contribution alone and its contribution given the other block, and the two values add up to the VAR's R^2_{oos} (black line). Table 1 of the letter, Panel C, instead drops lag t and uses lags $t-1$ to $t-5$. All forecasts follow the protocol of Table S1 and are evaluated on common dates.

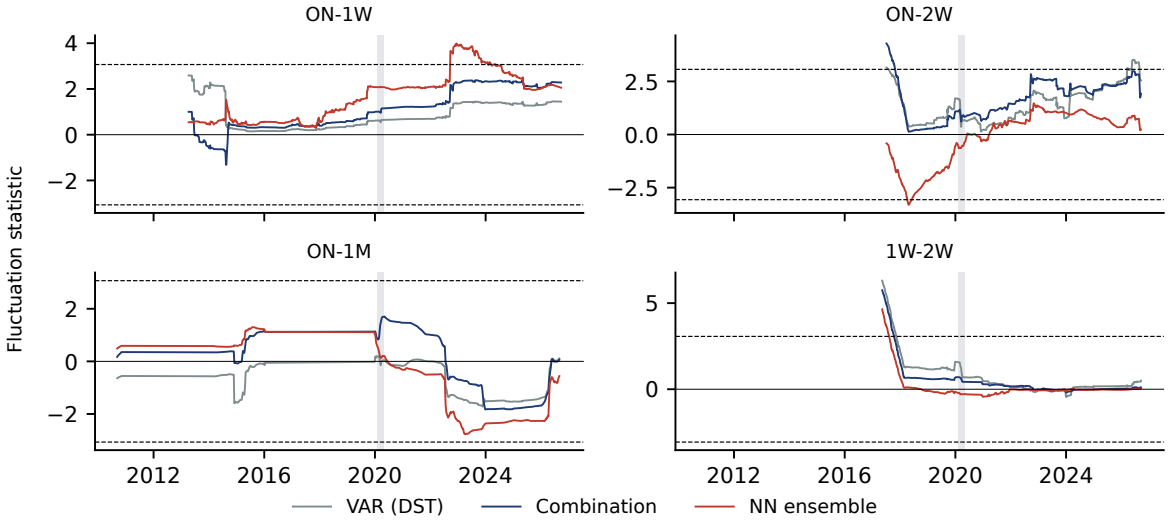


Figure S8: **Local Forecast Gains over Time.** Fluctuation statistic of Giacomini and Rossi (2010): standardised rolling mean of the difference between the squared errors of the EH forecast and those of the VAR, the combination of all forecasts, and the network ensemble, over a window of 30% of the out-of-sample period. Values above the upper dashed line indicate significantly smaller squared errors than the EH forecast at the 5% level. For nested forecasts such as the VAR and the EH forecast, the test is conservative, because the estimation noise of the larger model raises its squared errors; over short windows it therefore has little power. Shaded areas: NBER recessions.

S7 From Asset Pricing to Predictive Regressions

This section links the restriction of the letter to term-structure models, quote noise, and the perfect-foresight bound. It adapts the motivating framework of Bianchi, Büchner, and Tamoni (2021) to simple ACT/360 repo rates and to the rolling strategy of Equation (S1).

Decomposition. Write $a_j = i_{t_j}^{(m)} d_j/360$ for the simple return of the j-th roll, with $t_0 = t$ and $\sum_{j=0}^{k-1} d_j = D_t$. Then $r_t = \sum_j a_j + \tilde{c}_t$, where $\tilde{c}_t = \prod_j (1 + a_j) - 1 - \sum_j a_j$ collects the compounding cross-products. Inserting this into Equation (S1) gives

$$xr_t^{(n,m)} = \sum_{j=0}^{k-1} \frac{d_j}{D_t}\, i_{t_j}^{(m)} - i_t^{(n)} + \kappa_t, \qquad \kappa_t = \tilde{c}_t\, \frac{360}{D_t}. \tag{S3}$$

The weights d_j/D_t are known at t from the calendar. The compounding term κ_t is of second order: for a constant rate i and k rolls of equal length, $\kappa_t \approx i^2 D_t(1 - 1/k)/720$, about 1 basis point a year for ON-1M at a rate of 5% and less for the shorter pairs. Dropping it corresponds to $\ln(1 + a_j) \approx a_j$ in log-yield frameworks (Campbell and Shiller, 1991). Define the term premium and the expectation error as

$$\pi_t = i_t^{(n)} - \sum_{j=0}^{k-1} \frac{d_j}{D_t}\, E_t\left[i_{t_j}^{(m)}\right] - E_t[\kappa_t], \qquad \eta_{t+n} = \sum_{j=0}^{k-1} \frac{d_j}{D_t}\left(i_{t_j}^{(m)} - E_t\left[i_{t_j}^{(m)}\right]\right) + \kappa_t - E_t[\kappa_t]. \tag{S4}$$

Then, exactly,

$$xr_t^{(n,m)} = -\pi_t + \eta_{t+n}, \qquad E_t[\eta_{t+n}] = 0. \tag{S5}$$

Up to $E_t[\kappa_t]$, π_t is the premium $c(n, m)$ in Equation (1) of Della Corte et al. (2008), with calendar weights d_j/D_t in place of $1/k$.

Let x_t be the state that summarises the information at t. Then

$$E_t\left[xr_t^{(n,m)}\right] = -\pi(x_t) \equiv g(x_t) \tag{S6}$$

for an unknown function g, as in Equation (2) of Bianchi et al. (2021). The EH with a constant premium is the restriction that g is constant (Equation (2) of the letter). By the law of iterated expectations, the restriction then also holds for any observed predictors z_t dated t or earlier: $E[xr_t^{(n,m)} \mid z_t] = -\pi^{(n,m)}$. The forecasting problem is to estimate g from z_t. Under the EH the optimal forecast is the constant $-\pi^{(n,m)}$, which the prevailing mean estimates in real time, and any forecast that beats it out of sample is evidence against the EH, whatever its functional form.

Why g may be nonlinear. Term-structure models restrict g. In a Gaussian affine model with constant prices of risk, g is constant up to the second-order terms above, and the EH holds. With essentially affine prices of risk (Duffee, 2002), g is linear in the state; when the observed rates span the state, g is linear in the rates. If, in addition, expectations of the short rate are linear in lagged rates, this is the case tested by Della Corte et al. (2008): the VAR forecast, an ordinary least squares regression of $xr_t^{(n,m)}$ on the lags, is its reduced form. The linear VAR is therefore the natural benchmark, and flexible learners its natural generalisation.

At the short end there are several reasons to expect departures from linearity. Near the zero lower bound the short rate is approximately the maximum of a shadow rate and the bound, so expected short rates are convex in the state. Under the EH this convexity cancels in $xr_t^{(n,m)}$, but it invalidates a linear model of expectations, which is DST's caveat; a premium that compresses near the bound makes g itself nonlinear. Policy rates move in steps on scheduled

FOMC dates and within a corridor, so the expected short rate over the life of the long repo depends on whether a meeting falls within it, a threshold effect. Regime switches act similarly. Month and quarter turns, settlement days, and year-ends raise overnight rates on known dates (Hamilton, 1996), so the premium depends on whether the roll spans such a date, an interaction of calendar and rate level. Bianchi et al. (2021) use the same mapping from Equation (S6) to motivate neural-network forecasts of one-year Treasury excess returns.

Quote noise. Let the observed quote be $\hat{i}_s^{(\ell)} = i_s^{(\ell)} + e_s^{(\ell)}$ for $\ell \in \{m, n\}$, where $e_s^{(\ell)}$ has mean zero, is independent of the true rates at all dates and independent over time; the two noises may be correlated on the same date. The measured excess return $\widehat{xr}_t$ uses the quotes in Equation (S1). The long leg is linear in the quote, so its noise enters exactly with coefficient $-(360/D_t)(D_t/360) = -1$. The rolled leg is linear in each a_j, so the noise of roll j enters with coefficient $(d_j/D_t)\prod_{q\neq j}(1+a_q)$, where the product differs from 1 by at most about $i^{(m)}D_t/360$. Setting this product to 1 and ignoring terms that are quadratic in the noise,

$$\widehat{xr}_t \approx xr_t^{(n,m)} - e_t^{(n)} + \frac{d_0}{D_t}\, e_t^{(m)} + \sum_{j=1}^{k-1} \frac{d_j}{D_t}\, e_{t_j}^{(m)}. \tag{S7}$$

The long-rate noise of day t thus enters one-for-one with the opposite sign. The short-rate noise of day t enters only through the first roll, with weight d_0/D_t. This weight is $1/k$ on average: for ON-1W it is 1/7 from Monday to Thursday and 3/7 on Fridays, 1/5 over a week without holidays; for 1W-2W it is 1/2. The noise of later rolls is dated after t and cannot be predicted at t.

The predictors z_t contain the same quotes $\hat{i}_t^{(m)}$ and $\hat{i}_t^{(n)}$. Under the EH,

$$E\left[\widehat{xr}_t \mid z_t\right] \approx -\pi^{(n,m)} - E\left[e_t^{(n)} \mid z_t\right] + \frac{d_0}{D_t}\, E\left[e_t^{(m)} \mid z_t\right], \tag{S8}$$

which varies with z_t. Because the true rates are persistent, a jump in the same-day quote relative to its lags is partly noise: in a linear projection, the coefficient of $e_t^{(n)}$ on $\Delta\hat{i}_t^{(n)}$, with $\sigma_n^2 = \mathrm{Var}(e_t^{(n)})$, is $\sigma_n^2/(\mathrm{Var}(\Delta i_t^{(n)}) + 2\sigma_n^2) > 0$. A high same-day long quote therefore predicts a low measured excess return, even though nothing can be earned from it. If the predictors z_t^- are dated $t-1$ to $t-5$, they are independent of the noise of day t and of all later noise, so $E[\widehat{xr}_t \mid z_t^-] = -\pi^{(n,m)}$ under the EH. Panel C of Table 1 of the letter applies this test. In population and if the noise is independent over time, the R^2 without the quotes of day t is approximately a lower bound on predictability that does not come from noise, because dropping day t also removes genuine information. The drop in R^2 is approximately an upper bound on the noise component if, in addition, the genuine and the noise-driven parts of the forecast are not negatively correlated. Serially correlated noise, for example quotes that adjust slowly, would instead let lagged quotes predict as well.

Economic value. Each strategy holds the position $w_t = (\bar{\sigma}_t/\widehat{\sigma}_t)\,\mathrm{sign}(\widehat{\mu}_t)$ in the rolled leg, where $\widehat{\mu}_t$ is the forecast of the excess return, $\widehat{\sigma}_t$ the volatility of past excess returns, and $\bar{\sigma}_t$ its average over the estimation window; only the sign depends on the forecast. With $s_t = \mathrm{sign}(\widehat{\mu}_t)$, the return over the long repo is

$$w_t\, xr_t^{(n,m)} = \frac{\bar{\sigma}_t}{\widehat{\sigma}_t}\left|xr_t^{(n,m)}\right|\left(2\,\mathbf{1}\left\{s_t = \mathrm{sign}\left(xr_t^{(n,m)}\right)\right\} - 1\right). \tag{S9}$$

The mean return thus depends on sign accuracy weighted by $|xr_t^{(n,m)}|$, not on squared forecast errors. A forecast can reduce squared errors on many small excess returns yet earn little if it misses the sign of the large ones. Perfect sign foresight sets the indicator to 1 on every date and

attains the largest return, $(\bar{\sigma}_t/\widehat{\sigma}_t)|xr_t^{(n,m)}|$, whatever g is and whatever information a forecast uses (Section S6). This bound needs no estimate of g, which is why it complements the horse race of Section S3. Computed on quoted rates, it also includes the noise terms of Equation (S7) and so overstates what could be earned at transaction prices.

S8 Macro-Financial Predictors

Table S21 lists the 100 monthly predictors that extend DST's information set in Table S20, grouped into financial indicators (fundamental, liquidity, risk, sentiment, technical) and macroeconomic indicators (consumption, orders, and inventories; housing; labour market; money and credit; output and income; prices; sentiment), similar to the macroeconomic panel of Ludvigson and Ng (2009). Each series is transformed to stationarity with a code fixed by its type, following Ludvigson and Ng (2009) and McCracken and Ng (2016): interest rates in first differences; spreads, implied volatilities, diffusion and sentiment indices, and series that are already rates of change in levels; real activity, orders, employment, claims, exchange rates, and asset prices in first differences of logs; housing starts, permits, and sales in logs; prices, wages, money, and credit in second differences of logs. Where the prescribed transformation leaves a series nonstationary by an augmented Dickey–Fuller test at the 5% level (swap and transatlantic spreads; housing starts, permits, and sales; and the homebuilder index; nine series), the series is differenced once more; the 30-day relative strength index and the monthly change of construction spending, bounded rates of change, stay in levels although the test rejects only at the 10% level. The transformation codes were fixed before the results of Table S20 were computed. Five series stop updating within the sample: the three swap spreads in October 2022, after the end of LIBOR-based quoting, the three-month LIBOR in September 2024, and required reserves, which are zero from March 2020; after these dates their differenced values are zero and carry no information. Consumer sentiment and confidence, ratios, unemployment duration, and required reserves enter in first differences; the BBB corporate index enters as the three-month log change supplied by Bloomberg. Macroeconomic series enter with a publication lag of three months, financial series at the end of the month without lag. The monthly values are carried forward to the daily forecast dates and standardised within each estimation window. The series are revised data, not real-time vintages; the publication lag limits but does not remove the resulting look-ahead.

Table S21: **Macro-Financial Predictors**

No.	Series	Bloomberg ticker	Transformation	Lag (months)
Financial Indicator - Fundamental				
1	S&P 500 Index PE Ratio	SPX Index	$\Delta \ln x_t$	0
2	US Federal Funds Effective Rate (continuous series)	FEDL01 Index	Δx_t	0
3	US Commercial Paper Placed Top 30 Day Discount	DCPB030D Index	Δx_t	0
4	US Generic Govt 3 Mth	USGG3M Index	Δx_t	0
5	US Generic Govt 2 Yr	USGG2YR Index	Δx_t	0
6	US Generic Govt 10 Yr	USGG10YR Index	Δx_t	0
7	Moodys Bond Indices Corporate BAA	MOODCBAA Index	Δx_t	0
8	CHFUSD Spot Exchange Rate - Price of 1 CHF in USD	CHFUSD Curncy	$\Delta \ln x_t$	0
9	JPYUSD Spot Exchange Rate - Price of 1 JPY in USD	JPYUSD Curncy	$\Delta \ln x_t$	0
10	GBPUSD Spot Exchange Rate - Price of 1 GBP in USD	GBPUSD Curncy	$\Delta \ln x_t$	0
11	CADUSD Spot Exchange Rate - Price of 1 CAD in USD	CADUSD Curncy	$\Delta \ln x_t$	0
12	Bloomberg Commodity Index	BCOM Index	$\Delta \ln x_t$	0
13	Bloomberg Industrial Metals Subindex	BCOMIN Index	$\Delta \ln x_t$	0
14	10y. Transatlantic Spread	USGG10YR − GDBR10	Δx_t	0

No.	Series	Bloomberg ticker	Transformation	Lag (months)
15	2y. Transatlantic Spread	USGG2YR − GDBR2	Δx_t	0
16	USD SWAP SPREAD SEMI 2YR	USSP2 Curncy	Δx_t	0
17	USD SWAP SPREAD SEMI 5YR	USSP5 Curncy	Δx_t	0
18	USD SWAP SPREAD SEMI 10Y	USSP10 Curncy	Δx_t	0
19	ICE BofA US Corp BBB	C0A4 Index	level (3-month log change)	0
20	ICE LIBOR USD 3 Month	US0003M Index	Δx_t	0
21	US Muni AAA / 10-Yr Spread	.MAAA10YB G Index	level	0
Financial Indicator - Liquidity				
22	Bloomberg United States Financial Conditions Index	BFCIUS Index	level	0
23	US Corporate Baa / 10-Yr Spread	.BAA10YB G Index	level	0
Financial Indicator - Risk				
24	ICE BofA MOVE Index	MOVE Index	level	0
25	WTI Put Implied Volatility	CL1 COMB Comdty	level	0
26	Chicago Board Options Exchange Volatility Index	VIX Index	level	0
Financial Indicator - Sentiment				
27	AAII US Investor Sentiment Bullish Readings	AAIIBULL Index	level	0
28	AAII US Investor Sentiment Bearish Readings	AAIIBEAR Index	level	0
29	Bloomberg New Highs Minus New Lows NYSE	NWHLNYHL Index	level	0
Financial Indicator - Technical				
30	UST Future - 10 Yr RSI 30 Days	TY1 COMB Comdty	level	0
Macro Indicator - Consumption, Orders and Inventories				
31	ISM Manufacturing PMI SA	NAPMPMI Index	level	3
32	ISM Manufacturing Report on Business Supplier Deliveries	NAPMSUPL Index	level	3
33	ISM Manufacturing Report on Business Inventories NSA	NAPMINV Index	level	3
34	US Manufacturers New Orders Consumer Goods SA	TMNOCONS Index	$\Delta \ln x_t$	3
35	US Manufactures New Orders Total NSA	DGNOMATL Index	$\Delta \ln x_t$	3
36	Capital Goods New Orders Nondefense Excluding Aircraft SA	CGNOXAIR Index	$\Delta \ln x_t$	3
37	US Durable Goods Unfilled Orders Total SA	DGUOTOT Index	$\Delta \ln x_t$	3
38	US Manufacturing & Trade Inventories Total SA	MTIBCHNG Index	level	3
39	US Manufacturers Inventories to Shipment Ratio Durable Goods Industries SA	MINSDRGD Index	Δx_t	3
40	Merchant Wholesalers Inventories Total Monthly % Change	MWINCHNG Index	level	3
41	US Durable Goods New Orders Industries MoM SA	DGNOCHNG Index	level	3
42	US Auto Sales Total Annualized SAAR	SAARTOTL Index	$\Delta \ln x_t$	3
43	Richmond Manufacturing Survey Current Manufacturing Composite SA	RCHSINDX Index	level	3
Macro Indicator - Housing				
44	US New Privately Owned Housing Units Started by Structure Total SAAR	NHSPSTOT Index	$\Delta \ln x_t$	3
45	Private Housing Authorized by Bldg Permits by Type Total SAAR	NHSPATOT Index	$\Delta \ln x_t$	3
46	Census Bureau US Construction Spending MoM SAAR	CNSTTMOM Index	level	3
47	US New One Family Houses Sold Annual Total SAAR	NHSLTOT Index	$\Delta \ln x_t$	3
48	National Association of Home Builders Market Index SA	USHBMIDX Index	Δx_t	3
Macro Indicator - Labor Market				
49	US Employment Civilian Labor Force Total in Labor Force SA Household Survey	USLFTOT Index	$\Delta \ln x_t$	3
50	US Unemployment Duration Average SA	USDUMEAN Index	Δx_t	3
51	US Unemployment Duration Less Than 5 Weeks SA	USDULSFV Index	$\Delta \ln x_t$	3
52	US Unemployment Duration 5 to 14 Weeks SA	USDUFVFR Index	$\Delta \ln x_t$	3

No.	Series	Bloomberg ticker	Transformation	Lag (months)
53	US Unemployment Duration Over 15 Weeks SA	USDUFIFT Index	$\Delta \ln x_t$	3
54	US Unemployment Duration 15 to 26 Weeks SA	USDUFITS Index	$\Delta \ln x_t$	3
55	US Unemployment Duration 27 Weeks and Over SA	USDUTWSV Index	$\Delta \ln x_t$	3
56	US Initial Jobless Claims 4 Week Moving Average SA	INJCJC4 Index	$\Delta \ln x_t$	3
57	US Employees on Nonfarm Payrolls Total Private SA	NFP P Index	$\Delta \ln x_t$	3
58	US Employees on Nonfarm Payrolls Mining Industry SA	USMMMINE Index	$\Delta \ln x_t$	3
59	US Employees on Nonfarm Payrolls Nondurable Goods Production Workers SA	USENPROD Index	$\Delta \ln x_t$	3
60	US Employees On Nonfarm Payrolls By Industry Retail Trade Total SA	USRTTOT Index	$\Delta \ln x_t$	3
61	US Employees On Nonfarm Payrolls Total Government SA	USEGTOT Index	$\Delta \ln x_t$	3
62	ISM Non-Manufacturing Report on Business Employment SA	NAPMNEMP Index	level	3
63	US Employees on Nonfarm Payrolls Total MoM Net Change SA	NFP TCH Index	level	3
64	US Employees on Nonfarm Payrolls Manufacturing Industry Monthly Net Change SA	USMMMNCH Index	level	3
65	US Initial Jobless Claims SA	INJCJC Index	$\Delta \ln x_t$	3
66	US Continuing Jobless Claims SA	INJCSP Index	$\Delta \ln x_t$	3
67	US Personal Income MoM SA	PITLCHNG Index	level	3
Macro Indicator - Money and Credit				
68	Federal Reserve United States Money Supply M1 SA	M1 Index	$\Delta^2 \ln x_t$	3
69	Federal Reserve United States Money Supply M2 SA	M2 Index	$\Delta^2 \ln x_t$	3
70	Monetary Base Total NSA	ARDIMONY Index	$\Delta^2 \ln x_t$	3
71	US Aggregate Reserves Depository Institutions Required NSA	ARDIRRNA Index	Δx_t	3
72	Federal Reserve Consumer Credit Nonrevolving Outstanding Pct Chg at Annual Rate	CCARNREV Index	level	3
73	Conference Board US Lagging Consumer Installment Credit to Personal Income Ratio	LAGICRED Index	Δx_t	3
74	Federal Reserve Consumer Credit Total Net Change SA	CICRTOT Index	level	3
Macro Indicator - Output and Income				
75	US Personal Consumption Expenditures Ex Food & Energy Deflator SA	PCE CORE Index	$\Delta^2 \ln x_t$	3
76	US Industrial Production SA	IP Index	$\Delta \ln x_t$	3
77	US Industrial Production Nondurable Consumer Goods SA	IPNDTOTL Index	$\Delta \ln x_t$	3
78	US Industrial Production Business Equipment SA	IPEQBUS Index	$\Delta \ln x_t$	3
79	US Industrial Production Materials Nondurable SA	INDMTOT Index	$\Delta \ln x_t$	3
80	US Industrial Production Special Aggregates Converted Fuel SA	IPTSFUEL Index	$\Delta \ln x_t$	3
81	Adjusted Retail & Food Services Sales SA Total Monthly % Change	RSTAMOM Index	level	3
82	Adjusted Retail Sales Less Autos and Gas Stations SA MoM Percent Change	RSTAXAG% Index	level	3
83	US Capacity Utilization % of Total Capacity MoM	CPTIMOM% Index	level	3
Macro Indicator - Prices				
84	US PPI Finished Goods SA	PPI INDX Index	$\Delta^2 \ln x_t$	3
85	US PPI Finished Goods Less Foods & Energy SA	PPI IXFE Index	$\Delta^2 \ln x_t$	3
86	US CPI Urban Consumers Apparel SA	CPSCTOT Index	$\Delta^2 \ln x_t$	3
87	US CPI Urban Consumers Medical Care SA	CPUMTOT Index	$\Delta^2 \ln x_t$	3
88	US CPI Urban Consumers Commodities Less Food & Energy NSA	CPRPCXFE Index	$\Delta^2 \ln x_t$	3

No.	Series	Bloomberg ticker	Transformation	Lag (months)
89	US CPI Urban Consumers Commodities Durables SA	CPCADUR Index	$\Delta^2 \ln x_t$	3
90	US CPI Urban Consumers Services SA	CPSSTOT Index	$\Delta^2 \ln x_t$	3
91	US CPI Urban Consumers Less Food & Energy SA	CPUPAXFE Index	$\Delta^2 \ln x_t$	3
92	US Avg Hourly Earnings Private Nonfarm Payrolls Goods Producing SA	USHEGPSA Index	$\Delta^2 \ln x_t$	3
93	US Avg Hourly Earnings Private Nonfarm Payrolls Construction SA	USHECONS Index	$\Delta^2 \ln x_t$	3
94	US Avg Hourly Earnings Private Nonfarm Payrolls Manufacturing SA	USHEMANS Index	$\Delta^2 \ln x_t$	3
95	BDI Baltic Exchange Dry Index	BDIY Index	$\Delta \ln x_t$	3
Macro Indicator - Sentiment				
96	Market News International Chicago Business Barometer SA	CHPMINDX Index	level	3
97	University of Michigan Consumer Sentiment Index	CONSSENT Index	Δx_t	3
98	Conference Board Consumer Confidence SA 1985=100	CONCCONF Index	Δx_t	3
99	Philadelphia Fed Business Outlook Survey Diffusion Index General Conditions	OUTFGAF Index	level	3
100	Conference Board US Leading Index MoM	LEI CHNG Index	level	3

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